



CASE BOOK 2007

MANAGEMENT CONSULTING ASSOCIATION

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Case 1: University Bank

BACKGROUND

Firm: Boston Consulting Group

Round: 2007 Summer, First

Content: Qualitative and quantitative

CASE QUESTION

In Mexico banks prefer not to lend money to hospitals and schools because if one of these institutions defaults the Bank worries that it will be seen as the one responsible for liquidating their assets. We have a client who currently owns several universities. He is evaluating whether an opportunity exists to start a bank that would lend money to his various universities. There are three questions that he wants us to solve:

- Is there an opportunity to lend money to universities?
- If an opportunity exists, how big is it?
- How should he structure the bank?

INTERVIEWER BRIEFING

Recommended approach: The purpose of this case is to test the interviewee's ability to determine whether it makes sense for the client to open a bank that serves universities. In particular, the interviewee should recognize that he/she needs to gather the necessary information to determine the market opportunity for such a bank. The interviewee should ask questions to gather the necessary information to determine the market size.

Key facts:

- Population in Mexico is:
 - o 0-15 years old: 7 million
 - o 15-30 years old: 6 million
 - o 30-45 years old: 5 million
- Total number of universities in Mexico: 200
- Population growth: close to zero (use zero for the sake of simplicity)

EXAMPLE DIALOGUE

Interviewer: *So, given the facts of the case, how would you consider going about analyzing this question for the client?*

Interviewee: I'd like to understand a few things to evaluate this decision. First, I'd like to understand the size of the market to see whether there is an opportunity or not. Next I

would like to analyze who are the competitors, if any. In particular, I am interested in the competitors because they may decide to offer similar loans to institutions, like universities. Finally, I would like to know what the university can leverage from its own operations to develop this idea, etc.

Interviewer: *That seems like a good starting point. Let's start by analyzing the market. How would you do that?*

Interviewee: Ok, I will start by analyzing how many people go to university per year. If the population grows at about 1% per year, this means that next year there could be more people attending universities than space available. I would use a guesstimate of the percent of population enrolled in universities to calculate the number of students currently enrolled across Mexico. For simplicity sake I will assume that all potential students attend universities in Mexico, excluding the percent that attends school abroad.

Age range	Population	% Enrolled in University (guesstimate)	Enrolled In University
0-15 years old	7 million	0%	0.0 million
15-30 years old	6 million	25%	1.5 million
30-45 years old	5 million	2%	0.1 million
			1.6 million

There are roughly 6 million Mexicans between the ages of 15 and 30, and I would assume that they are evenly distributed. That is, there are 400,000 people in each year of age (e.g. 400,000 people who are 15 years old, 400,000 people who are 16 years old, and so on until the age of 30). If I assume that 25% of the people between the ages of 15 and 30 go to university, that means we have 1.5 million students between the ages of 15 and 30.

Interviewer: *Those assumptions seem to be very good ones. What else would you need to consider?*

Interviewee: Ok, I think that I will need to figure out how many schools there are in Mexico and how many students are enrolled in each school.

Interviewer: *Sure, our client has told us that there are 200 universities in Mexico and we can assume there they are all at full capacity.*

Interviewee: Ok, so 1.6M divided by 200 means that there are 8,000 students at each school. I might think that this seems like a high number. In the United States we have part-time students and full-time students. If this is similar in Mexico, the universities would likely be able to accommodate more students without any investment.

Interviewer: *No, they can't do that as the law prohibits it.*

Interviewee: Okay, so if the population grows at 1% per year and there are no new people going to university for other reasons (e.g. better economy) this means that in any given year we will have 1.6M multiplied by 1% new students. We would have 16,000 incremental students. Given that we know the schools are at full capacity, we would need two new schools.

Interviewer: *Okay, so now let's focus on how would you segment the schools in order to think about the credit scoring they may have?*

Interviewee: I would segment them based on:

- New school or just new branch to an existing school. I believe that it is less risky to lend to a company that already has a business in place.
- Flow of people to school: it's much more secure to lend to a school that has plenty of students in each year than to one that has the same number but distributed in a different way (e.g. most of the people finishing their careers and almost nobody in the first or second year).
- Geography: We are speaking in averages. There may be areas with no population growth. There may be an uneven dispersion.

Interviewer: *How would you calculate how much money a typical school would need?*

Interviewee: There are two different streams of costs we have to consider. On the one side the cost of building a new school – including buying the land, the construction, etc. – and the funds needed until the company reaches a state in which it has a positive free cash flow.

Interviewer: *Great. Tell me the number, or at least how to get it.*

Interviewee: The better is to rely on our client experience. He has 15 schools so there are some experiences we can leverage there?

Interviewer: *Let's move on. How would you fund this project?*

Interviewee: I will think about a couple different ideas such as:

- Unilateral organizations (e.g. World Bank, etc.)
- The Government as it may need to help some private schools that focus on careers that are not available through public university.
- Banks: They may not be willing to do this on their own but may want to partner with someone that knows about this industry.

Interviewer: *Ok, fair enough. Our client is asking you to summarize the situation. What would be your recommendation?*

Interviewee: Can I take a moment to recap all the findings?

Interviewer: *Sure.*

Interviewee: I will tell the client that it seems to be a pretty interesting opportunity to provide capital to build new schools which would be at least two per year given the growth in the population.

Case 2: Mine in Chile

BACKGROUND

Firm: Boston Consulting Group

Round: 2007 Summer, First

Content: Operations

CASE QUESTION

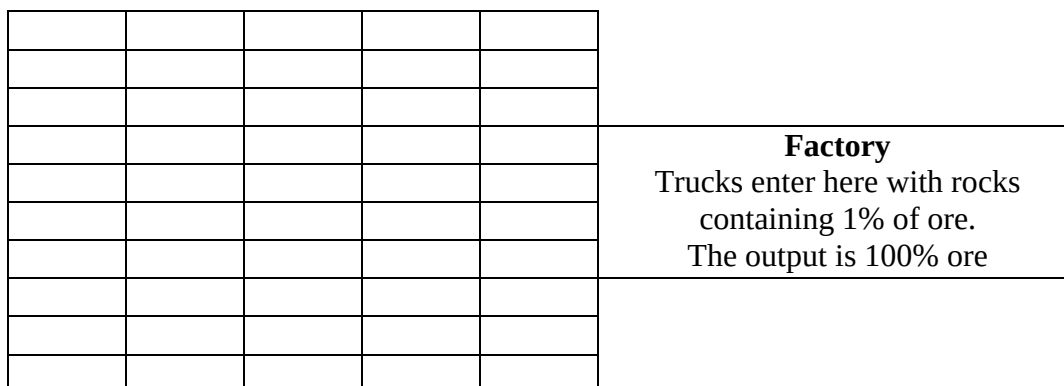
Our client is a company which is going to exploit a mine that is expected to be full of a very unique ore, and has asked us in which order his company should extract the ore from the field. The ore will then be sent to a factory that his company already owns.

INTERVIEWER BRIEFING

Recommended approach: The interviewer is testing whether the candidate can identify a useful algorithm to determine the best way to extract ore from this field. The candidate should not initially waste any time developing one idea too fully; instead he/she should come up with a few reasonable suggestions and test them for validity with the interviewer. The candidate should, like in most cases, be aware that he/she is missing critical information and should identify what he/she needs to solve the case and then ask the interviewer for this data.

Key facts:

- The field is divided in cells of equal size. Once you blow one cell there is no way the truck can pass through it to go to the factory.
- This is a plan of the plant:



- Each cell has different ore content and therefore each cell has a different cost of extraction.
- No growth in population (for sake of simplicity).

EXAMPLE DIALOGUE

Interviewer: *So, tell me how we might approach this problem for our client?*

Interviewee: First, I would like make sure that I fully understand our client's needs. Let me repeat the key issues... Our client wants us to provide him with a recommendation on what approach his team should take to extract the ore in order to maximize profits.

Interviewer: *Yes, that pretty much sums it up. So, what's next?*

Interviewee: I can imagine a number of different ways of prioritizing which cells to take out first, such as:

- Profit: Price – Cost.
- Demand: Outlook of the price of the different types of ore. If the price is low (perhaps because demand today is lower), but we believe that this same ore will be sold for much more in the future (perhaps due to an increase in demand), it could make sense to begin extracting lower concentration cells.
I am imagining that this could be similar to the oil industry. Think for example about the price of oil. The oil is in the field anyway so it's better to exploit the places which the least of it when the price in the market is at a low level. Unless, of course, we can store the ore for long periods of time without a large opportunity cost by tying up capital and space.
- Capacity of the plant: Maybe there is a minimum level we need to provide to the plant as input. Maybe there is a maximum... this could determine how much we want to extract in any given time period, especially, if there are penalties for operating outside of this min-max range.
- Operational issues: For example, if I dig a hole in the ground and the truck cannot pass through that cell, it may be wise to start at the cells that are further from the factory so that I do not block off routes.

Interviewer: *Ok, let's assume that the points that you have raised are all reasonable. This is a complex business and I was pleased to hear you touch on many of the important points. But, first, let's focus on profits. How would you define profit?*

Interviewee: Ok, in this case profits would be (price of a ton of ore) * (expected content of ore in cell) – (the cost of extraction for that cell). I will then rank each of the cells based on its overall profit contribution and determine my extraction path from there. Of course, we need to consider that if we extract from one highly profitable cell, but it cuts

off access to a lesser, but still profitable, cell, this is a cost of doing business under this assumption and should reduce my overall expected profits.

Interviewer: *Let's assume that we map the field and this is what we get:*

Cell 1 x = 1.0 C = 3					
	Cell 2 x = 1.3 C = 10				Factory
				Cell 3 x = 1.1 C = 5	

x = tons of ore in that cell

C = cost of extracting in that cell

Interviewee: Do we have any information on today's price of ore?

Interviewer: *Yes, our client expects to get \$10 dollars per ton of ore.*

Interviewee: So, for Cell 1 the profit will be: $1 \cdot 10 - 3 = 7$. For Cell 2 would be: $1.3 \cdot 10 - 10 = 3$. For Cell 3 would be: $1.1 \cdot 10 - 5 = 6$. So I will start for Cell 1, then Cell 3 and finally exploit Cell 2.

Interviewer: *Ok. What if price changes? Does it change your decision?*

Interviewee: Yes.

Interviewer: *Why?*

Interviewee: Let's say price is now \$30 per ton. For Cell 1 the profit will be: $1 \cdot 30 - 3 = 27$. For Cell 2: $1.3 \cdot 30 - 10 = 29$. For Cell 3: $1.1 \cdot 30 - 5 = 28$. So I would recommend extracting Cell 2, then Cell 3 and finally Cell 1. This is based on the fact that the costs appear to be fixed and do not change despite the overall price fluctuations that we are discussing.

Interviewer: Let's move on. Our client has told us that per ton P&L statement is as follows...wait, perhaps you could tell me how you would construct it?

Interviewee: I would try to find out the average price, the average cost of goods sold and then assign some part of the SG&A costs.

Interviewer: Ok. Here are the figures that our client provided. What is the company's break-even point?

Item	\$ per ton
Revenues	1.0
COGS	0.8
SG&A	0.6
Profit	-0.4

Interviewee: Ok. The company has \$0.2 gross margin per ton of ore, before we factor in SG&A costs. So, if I divide SG&A by the gross margin I will find how much they have to produce. In this case, $\$0.6 / \$0.2 = 3$, which means they have to produce 3 times more to cover fixed costs.

Interviewer: How the same P&L would look like?

Interviewee: Basically, we have to divide the SG&A by 3. We are in essence spreading the fixed SG&A costs across more units. However, we would need to make sure that SG&A does not increase with volume of ore extracted.

Item	\$ per ton
Revenues	1.0
COGS	0.8
SG&A	0.2
Profit	0.0

Interviewer: Okay, can you summarize what we have identified?

Interviewee: Sure. Our client was looking for a framework to identify the order in which he should extract cells of ore from a field. We chose to focus on profitability as the main driver. As mentioned, but not discussed in depth, profitability includes the core components of price per ton of ore, cost to extract the cell (which appears to be fixed), but also the cost of not being able to access a cell for extraction. Once we have

calculated the profits per cell our client can draw out a map to understand which cells to extract and when in order to maximize his P&L. Next we discussed the company's P&L and identified that SF&A costs are relatively significant and appear to be fixed. This means that we would want to extract more ore in order to spread these costs across production and increase overall profitability.

Case 3: UPS in Italy

BACKGROUND

Firm: McKinsey & Company

Round: 2007 Summer, First

Content: Quantitative

CASE QUESTION

The CEO of a startup in a small village in Italy has hired McKinsey to help them decide how many trucks to lease. There are different models available, but our client has been told that he/she will need to have a consistent fleet (they can only lease one model type) and so we will also need to identify what model he/she should lease. This company provides the local delivery of packages sent to this village through UPS next-day-delivery service.

Let me provide a quick overview of how the company operates: (i) They receive every package at 5pm from UPS, (ii) a bunch of people then sort the packages and (iii) load them on a truck where they are stored overnight, and (iv) then deliver them starting at 9am for 10 hours. How would you suggest approaching the client's problem?

INTERVIEWER BRIEFING

Recommended approach: This case tests a candidate's ability to analyze how many packages must be delivered and to see if the bottleneck is the time or the truck size. Not all information is provided up front to the candidate; he/she should be aware of this and must identify additional data that will allow him/her to solve the case.

Key facts:

- Packages delivered per day: 1,000.
- Dimension of package (envelope) is 1x1x1.
- Operates five days a week.
- It takes 8 minutes, on average, to deliver a single package and to be ready for the next one ("assume they deliver one every 8 minutes").
- Truck A costs \$150 per day and its dimensions are 3*4*5.
- Truck B costs \$140 per day and its dimensions are 9*2*1.
- Truck C costs \$130 per day and its dimensions are double the size of truck A.
- Drivers, fuel, etc. are not considered and do not make a material difference to the analysis (for sake of simplicity).
- The case will include additional questions not addressed in the initial scope.

EXAMPLE DIALOGUE

Interviewer: *So, how would you go about analyzing this problem?*

Interviewee: I'd like to understand a few things to evaluate this decision. First, I would like to start by analyzing the demand. I would like to know how many packages we have to deliver and how long, on average, it would take us to deliver a single package. Then, I would like to analyze the numbers in the context of the three truck models our client can lease.

Interviewer: *Ok. We can satisfy a demand of 1,000 packages / day and it takes 8 minutes on average to deliver each one.*

Interviewee: So, $1,000 \text{ packages} * 8 \text{ minutes} / 60 \text{ minutes} * 10 \text{ hours} = 13.3 \text{ trucks}$. So we need at least 14 trucks. I would like to think about the leases we can consider.

Interviewer: *Ok. Let me show you the information we received from the client:*

Truck	Cost per day	Dimensions
A	\$150	3*4*5
B	\$40	9*2*1
C	\$130	6*8*10

*Truck A cost \$150 per day and its dimensions (for the packages) are 3*4*5. Oh, by the way, you did not ask about the average size of an envelope, but our client has told us that the average size is 1*1*1.*

Interviewee: Ok, so when assessing Truck A we multiply 1,000 (total packages for all trucks) by 1*1*1 (average package size) and divide by 3*4*5. We will know how many trucks we would need of the Truck A model. My calculations show 16.6 which would mean we would need 17 of this type of truck. As 17 is more than the time constraint of 14 truck to ensure on-time delivery, we stick with 17.

So, this is the result of performing this analysis for each type of truck:

Truck	Cost per day	Dimensions	# trucks (rounded)	# trucks (minimum)	Total cost per day
A	\$150	3*4*5	17	17	\$2,550
B	\$40	9*2*1	56	56	\$2,240
C	\$130	6*8*10	3	14	\$1,820

Now, from a pure financial analysis, I would recommend leasing 14 trucks of the C Model because it will allow our client to minimize the cost while ensuring on-time

delivery (customer satisfaction). On the other hand, we also might consider that there would be plenty of room for delivering other things if they can figure out how in the future.

Interviewer: *Ok, it seems a good idea. Let's move on. Now imagine 6 months have passed and your recommendation was pretty successful. Now the CEO want us to investigate any potential risks that he/she should be assessing/considering.*

Interviewee: Can I take a minute to organize my thoughts?

Interviewer: *Ok.*

Interviewee: So, I would like to go over this problem by analyzing both internal and external factors. Here there is a list of the things I would think about:

Internal	External
▪ Need for extra drivers (e.g.: people get sick) – do we have enough employees ▪ Unionized drivers may shift labor cost up in the future ▪ Need to lease more trucks because trucks can break down causing late delivery ▪ Insurance costs ▪ Extra fine tickets than forecasted because drivers want to deliver on time ▪ Etc.	▪ Only one supplier (UPS) – we are captive to UPS ▪ Adoption of new technology (e-mail) might reduce the need for sending packages ▪ Government regulation ▪ New competition in the city – there are no real barriers to entry, since UPS would likely partner with any carrier who can deliver on customer service metrics at a cheaper cost ▪ No association with our brand, thus our supplier can switch to our competitors or start its own operation ▪ Etc.

Obviously, I could analyze the sorting operation more to make a more profound (exhaustive) analysis but that was not covered on our initial discussion.

Interviewer: *Don't go that direction. Let's think of another scenario. Now we have to investigate sources for profit growth for this company with one restriction, we can neither add new truck leases nor change the existing ones.*

Interviewee: Ok. Let me think, about increasing revenues:

- Extend hours: the trucks are already paid for the day, if we extend the delivery time after 7pm we can deliver more of UPS or from other companies, even local companies. That would go (impact) directly to profits.
- Different packages: we may recommend to UPS to sell different (more robust) packages to some clients and get part of it.
- Pick packages: every time we leave a package we make space to pick a package and deliver it to another part of the village or to give it back to UPS to send it to another place
- Get contract with a new operator: see whether we can deliver stuff to other company who is in the delivery business but does not compete directly with UPS. Thought we can not add new trucks we can think about utilization of current trucks
- Advertisement: are the trucks painted with UPS logos? We can sell advertisement to them or to other companies. Those trucks are all day in the street.
- Insurance: offer insurance of packages to clients.

Interviewer: *What else?*

Interviewee: We can also think about streamlining operations (reducing costs). I would think about them in terms of fixed and variable costs.

- Evaluate the route of each truck to reduce time or usage of gas (fuel)
- Improve technology usage in the sorting and loading packages. May reduce number of people at the factory
- Re-negotiate leasing terms for trucks
- Move warehouse to a cheaper place

Interviewer: *Ok, we are out of time, but thank you for the list of business challenges that the client should assess, I am sure that some of these will lead to very interesting discussions.*

Case 4: Refrigerators in India

BACKGROUND

Firm: McKinsey & Company

Round: 2007 Summer, Second

Content: Quantitative and qualitative

CASE QUESTION

A producer of high-end refrigerators in India has hired McKinsey to help him solve a profitability problem. Two years ago the company had a record year in terms of growth. Though they were correct in understanding the size of the market and how it would grow, their gross profit decreased by approximately 70% and the CEO is very worried about the situation. The client has asked us to identify what happened and what can be done to restore profitability?

INTERVIEWER BRIEFING

Recommended approach: This is a typical profitability question. A good way to go over it is to break down the profit equation into the different concepts, build hypothesis, and test them.

Key facts:

- 800 million people in India
- Two years ago units sales in India were 5 million and last year 6 million
- The client's company builds and sells high-end units

EXAMPLE DIALOGUE

Interviewee: Can I take a minute to think about this problem?

Interviewer: Yes.

Interviewee: Ok. I would start by breaking down the problem into revenues and costs in order to identify the potential drivers of the drop in profitability.

Interviewer: Great. Let's start with the costs.

Interviewee: Ok. Given that we know that this market has been growing quickly, I would like to know whether our client has decided to build a new factory to keep up with the

growth. If that was the case, depreciation and amortization and interest on debt (if it was paid by issuing debt) could be putting pressure on the profit of the firm.

Interviewer: *You are definitively right in your analysis, but I have to tell you that although they have built a new plant, there is no problem on the cost side. I can tell you that since we have just finished another engagement in which we streamlined all his operations.*

Interviewee: So, now I would like to understand a little more about the revenue side of the profit equation. Perhaps I could learn a little more about the market. I'm most interested in having some assumptions to determine the size of the market.

Interviewer: *Sure. What do you want? If you ask for specific information I may have it.*

Interviewee: Ok, I would like to know the population of India, the number of households and what kind of refrigerators the client produces.

Interviewer: *Assume 800 million people and... I am not sure about households... guess something.*

Interviewee: I would say that there might be 5 people per household, which leads to 100 million households. Based on my experience I will say people change their refrigerator once every 5 to 7 years.

Interviewer: *Sounds reasonable, let's say 7 years.*

Interviewee: If I had the number of units sold in the market last year I would be able to understand how many people actually have refrigerators at home in India. Do we have any information around the percent of households that have a refrigerator?

Interviewer: *Last year sales were 6 million units and one year before they were 5 million units.*

Interviewee: So,

Item	Quantity
Population	800,000,000
# per household	5
Households	160,000,000

Income	# sold	Years between change	Market size
Two years ago	5,000,000	7	35,000,000
Last year	6,000,000	7	42,000,000

Interviewer: *You are saying that the number of refrigerators for home-use was 35 millions two years ago and 42 millions last year?*

Interviewee: Yes. However, I might question whether the 7 million growth of last year could have been in a segment not served by your client.

Interviewer: *How do you know that?*

Interviewee: Well, we concluded that there are 160 million households and from our estimation we get that only $42/160 = 26\%$ have a refrigerator. On the other hand you mentioned that they produce high-end which may signal that the growth is coming from the low income part of the society, for which your product may not be priced at a competitive level or even the product not designed for their needs.

Interviewer: *You are completely right. What we should do?*

Interviewee: There are several options. Our client can try to use the extra capacity they have in the new plant to focus on building lower-end refrigerators, leveraging their supplier relationships, dealers, etc. However, we would need to consider how to brand these refrigerators so as to not detract from the quality associated with the higher end of the market. They can think about exporting the extra capacity to other countries where their product offering is competitive and there is market for growth, they can start offering financing for this particular segment...

Interviewer: *Fair enough, let's imagine that the marketing director came last night and proposed to reduce price by 20% to appeal to this segment, which he thinks will increase sales by 30%. Should we support him?*

Interviewee: Is that all the information we have?

Interviewer: Yes.

Interviewee: Let me think. Let's denote P = Price, C = Cost of goods sold, Q = Quantity. The general formula to make this decision should be $P*Q - C*Q > 0$. In this case:

$$80\%*P*130\%*Q - C*130\%*Q > 0$$

$$104\%*P*Q - C*130\%*Q > 0$$

Break-even point at:

$$104\%*P*Q = C*130\%*Q$$

$$104\%*P = C*130\%$$

$$104\%/130\% = C/P$$

$$80\% = C/P$$

COGS of 80% over price does not seem to be completely out of what we would expect in a manufacturing company. I would worry about COGS being just 25%, which will signal that reducing price by 20% is not a good idea to attain 30% more sales.

Interviewer: Thanks.

Case 5: Mexican Dishes

BACKGROUND

Firm: A.T. Kearney

Round: 2007 Summer, First

Content: Quantitative

CASE QUESTION

A friend of mine has asked me to invest in his company, which makes dishes for the Mexican market. He has mentioned that he is going to use my investment to purchase a new manufacturing machine that he expects will allow him to double production. Should I invest in his venture?

INTERVIEWER BRIEFING

Recommended approach: Rather than jumping directly to analyze in depth the market, the competition and/or the company's capabilities, it is recommended that you use a dialogue to seek out more information. When mentioning the themes you would like to cover explain your reasoning behind each one.

Key facts:

- 110 million people live in Mexico. Assume 100 million for sake of simplicity.
- Population growth can be approximated to be 1% per annum.
- Income distribution: 5% of population top, 45% middle class, 50% bottom.

EXAMPLE DIALOGUE

Interviewer: *So, should I invest or not?*

Interviewee: Ok, I would like to analyze three key areas that I expect will help me understand whether this is a good opportunity or not. I believe it is important to first analyze the market, focusing on its size, growth potential and trends in terms of product design, then understand who are my friend's key competitors and finally understand what this company can leverage from its existing operations once the new machine is in place.

Interviewer: *Good idea. Let's focus on the market for a moment. How would you go about sizing it?*

Interviewee: There are a number of ways to do so. We can do some research and talk to your potential partner to get some information. Alternatively, we can visit some of the

store or chains that sell that kind of product to build a market size from the bottoms up. Finally, we can estimate that number based on the population, etc.

Interviewer: *Ok, let's do the estimation based on the population. In Mexico there are 100 million people. What else do you need?*

Interviewee: Do you have the income distribution pyramid of Mexico? I believe that income will play a role because households with higher incomes may purchase higher-end dishes or may purchase different quantities than households at the other end of the income distribution. I would also like to get a sense of the replacement rate, if not given I can estimate.... And, whether there are any key substitutes that I can expect in the market in the next few years, to ensure that my estimates are fair.

Interviewer: *Assume income distribution is 5% of population top, 45% middle class and 50% at the bottom. And, you can assume a 15% per year replacement rate.*

Interviewee: Ok. Let me quickly use these numbers to analyze the size of the market. I will calculate first “the stock”, which is the size of the market given by the replacement of dishes. Then I will calculate “the flow”, which is the size of the new market.

Item	Quantity
Population	100,000,000
# per household	4
Households	25,000,000

Income	# of different Sets	Total # of sets	Total # of sets (replaced)
High	3	3,750,000	562,500
Medium	2	22,500,000	3,375,000
Low	1	12,500,000	1,875,000

Interviewer: *But... Are you sure that we can address the whole market?*

Interviewee: No, in fact this was only to calculate the full market size. I still wanted to ask you which part of the market this company targets, is it within one income class?

Interviewer: *Okay, they target the medium segment, which by your calculations is of about 3.4 million sets per year, among medium income households.*

Interviewee: Great. However, I would still want to understand the projected market growth. If the population growth at 1% per year we can assume that there are about 250 thousands (25 millions * 1%) new houses per year, which 45% are of medium segment and they have to equip themselves with 2 sets... So there is an extra 225 thousands more dish sets demanded per year and a replacement rate of 15% on these new dishes. So, the market size is approximately 3.6 millions sets per year.

Item	Quantity
Population	100,000,000
Growth	1%
# per household	4
Households	250,000

Income	# of different Sets	Total # of sets (bought)
High	3	37,500
Medium	2	225,000
Low	1	125,000

Interviewer: *On the ballpark, my estimation was that it was around 4 million.*

Interviewee: Anyway, we are still missing other segments on the markets such as restaurants, hotels, offices, etc.

Interviewer (interrupts): *Don't worry about that. Let's continue with this.*

Interviewee: Okay, so now, we have to think whether it is convenient for you to enter the partnership. To do so I would like to do a break-even analysis of buying the machine and figure out the amount of sets that we have to sell...

[The interviewer interrupts again, since time is running low. Remember to practice good time management in interviews and take cues from the interviewer]

Interviewer: *Assume that we have already done so and we figure out that we have to produce at capacity (150,000 sets per year). We don't have much time. What would your recommendation be?*

Interviewee: So, given that the offer was to take the money to double production it means that you will go from 150,000 to 300,000. As this estimate represents less than 10% of the available market, his proposal at first glance seems feasible. However, I would want to know our current share versus those of our competitors in this segment. If a competitor already had 70% of the market, I might be more cautious about entering into this venture. Furthermore, if I had more time I would focus on what can be leverage from the current cost structure to assure it is a profitable proposition.

Case 6: Centralized Distribution

BACKGROUND

Firm: A.T. Kearney

Round: 2007 Summer, First

Content: Qualitative

CASE QUESTION

I was on the golf course the other day with a friend of mine who owns a chain of supermarkets and he posed a question that is still on my mind. His question was, why should he continue to centralize his company's sourcing rather than split it to procurement and distribution at the individual store level?

INTERVIEWER BRIEFING

Recommended approach: This is a pure conceptual case in which you have to come up with a number of reasons to go one way or the other. Be creative in the way you think about it, the case is open ended, but look for guidance from the interviewer if you start to go too far a field.

Key facts:

- N/A

EXAMPLE DIALOGUE

Interviewee: Let me take some time to think about how to go over this problem.

Interviewer: *Sure, take your time. [...] So, why should he centralize sourcing and distribution?*

Interviewee: In order to come with an answer I would like to go over the pros and cons of centralizing and then weight them in order to come with an answer.

Interviewer: *Fine with me.*

Interviewee:

Pros	Cons
▪ Ability to “squeeze” suppliers by aggregating demand ▪ Flexibility to change what to deliver to individual stores based on daily/weekly demand changes ▪ Economies of scale in warehouse size, operation, etc. ▪ Reduce cost of warehouse by having it in a low cost area and not in the store, which I assume is a high cost area ▪ Local people (at the store level) are not trained to negotiate. Having them at the store level is an extra cost ▪ Etc.	▪ Local people can understand better the necessities (e.g.: because of format, size, place) for some (niche) categories ▪ If distributed directly to the store the cost of distribution is paid by the supplier ▪ There may be some products that are not bought at the centralized (HQ) level because of limited size order. Managing at the store level make it viable ▪ Etc.

Interviewer: All good reasons... but I have talked to my friend about many of those things and he keeps telling me that there may be something else. What could it be?

Interviewee: So, we have covered most things and we are still not finding the reason...

Interviewer: Yes.

Interviewee: Looking at the ideas that I put in the pros vs. cons comparison, it could be similar to the notion of some products not being bought at the HQ because of some quantities asked for an individual store. What if the supplier refuses to ship to individual stores because of limited volume? He may not have the trucks to do so in such a small volume or he may be unwilling to do so because is not profitable to do in that way.

Interviewer: You got to the point. After a couple of hours in the golf course with my friend I came to the same conclusion, which later he validated with his team as the main reason.

Case 7: Health Care in Canada

BACKGROUND

Firm: McKinsey and Company
Round: 2007 Summer, First
Content: Qualitative and quantitative

CASE QUESTION

Our client is a Canadian pharmaceutical company. Their first drug has just been approved by what is equivalent to the United States' FDA and is an asthma treatment. The client has hired McKinsey to help them assess what is the right price for this new treatment and they will also need McKinsey's help with completing an application to the government for price approval. The second step is necessary because, as you may know, Canada's government subsidizes healthcare costs and we will discuss this in greater detail later in the case.

INTERVIEWER BRIEFING

Interviewees should consider what information will be necessary to address this case and ask for data as they see fit.

- Research and Development costs for this drug are estimated to be \$5 billion.
- Beyond R&D, marketing is the largest cost for a pharmaceutical. The interviewer, however, will ask the interviewee to assume that marketing costs are \$0 and there are no variable costs.
- There are three segments to this market:
 1. **Basic** – 2% of the population – have a periodic asthma attack, use 1 inhaler/month
 2. **Serious** – 2% of the population – use 1 inhaler/week
 3. **Acute** – 1% of the population – use 1 inhaler/week, but attack sometimes results in hospitalization and death.
- The client's new treatment is classified as preventative. It is a pill that must be taken every day.
- Regular inhalers cost \$10 each. This is the common treatment for asthma.
- Acute patients that result in hospitalization spend 1 night/year in the hospital at \$1000 and the rate of death is 1%.

EXAMPLE DIALOGUE

Interviewer: *First, let's discuss the possible ways of framing the client's first question – how should it price this new drug?*

Interviewee: Well, there are three different methods I can think of:

- Cost based pricing – set the drug price at cost and add a percentage markup
- Value based pricing – set it at what customers are willing to pay
- Determine Minimum and Maximum prices

Interviewer: *Great, now, go through the steps to solve this problem and provide a price to the client.*

Interviewee: Okay, so let's try cost based pricing. I would assume that the company has both fixed costs and variable costs. Have we gathered any information from the client about its cost structure?

Interviewer: *Well, what do you think would be its largest costs?*

Interviewee: Given what I know about the pharma industry and the extensive R&D for drugs, I would guess Research and Development.

Interviewer: *Good, R&D costs are \$5 billion. What would you guess are some other key categories of costs?*

Interviewee: Marketing is usually a big cost.

Interviewer: *Right. For this case, let's assume marketing is \$0 and there are no variable costs. [Where possible, the interviewee could have tried to anticipate this chain of questions and suggested, without being asked, the key categories of costs that might be relevant].*

Interviewee: Next I would want to size the Canadian market for this drug, assuming that we will only sell in Canada. To do this, the population of Canada is approximately 30 million people. We would now need to estimate the percentage that is asthmatics?

Interviewer: *There are 3 segments to this market:*

Basic – 2% of the population – have a periodic asthma attack, use 1 inhaler/month.

Serious – 2% of the population – use 1 inhaler/week.

Acute – 1% of the population – use 1 inhaler/week, but attack sometimes results in hospitalization and death.

Also, the treatment is preventative. It is a pill that must be taken every day.

Interviewee: Okay, so let's assume every segment will use this drug. $30 \text{ million} * 5\% = 1.5 \text{ million people}$. However, I would expect that there would be some barriers to switching and not all potential users will switch from inhalers.

Interviewer: *Okay, so now how do you set the price?*

Interviewee: I want to determine the price to break-even. For a pharma, I think 5 years is acceptable.

To break even in 5 years: $1.5 \text{ million} * \text{Revenue} = 5 \text{ billion (R\&D)}$
 $\text{Revenue} = \$3,300$

Use 350 days/year for simplicity, one pill every day over 5 years = $350 * 5 = 1,750 \text{ pills}$

$\$3,300 / 1,750 \text{ pills} = \text{approximately } \$2/\text{pill}$

Interviewer: *So do you think this price should be the minimum or the maximum?*

Interviewee: This is the minimum because it is the break-even price.

Interviewer: *Okay, let's consider the second half of the question. As I mentioned, the Canadian government subsidized medical costs, let's for the sake of this case say that it pays back its citizens for medical treatments. How do you determine how to set the price so that the government will agree to pay it?*

Interviewee: Assuming that the government is paying for the current treatment, I would want to know their current spend and determine what the difference is between that and the new drug. Do you know how much the government is currently paying patients for their inhalers?

Interviewer: *The cost of inhalers is \$10 each. Also, acute patients that result in hospitalization spend 1 night/year in the hospital at \$1,000 and the rate of death is 1%.*

Interviewee: So I want to determine how much the government is currently spending on inhalers.

Basic: $\$120/\text{year}$ (1 inhaler a month), 600,000 people, \$72 million total
Serious: $\$520/\text{year}$ (1 inhaler a week), 600,000 people, \$312 million total
Acute: $\$520/\text{year}$ (1 inhaler a week), 300,000 people, \$156 million total

Total government is spending on inhalers: \$540 million/year.

Now I want to determine how much government is spending on hospitalizations.

$\$1,000 * 300,000 \text{ people} = \300 million/year

Total government spend = $\$540 \text{ million} + \$300 \text{ million} = \$840 \text{ million}$

Over five years: $\$840 \text{ million} * 5 = \4.2 billion

So with the new drug, we calculated the break-even at five years to cover \$5 billion in R&D. With inhalers, in five years, the government is spending \$4.2 billion.

Interviewer: *Good, so with this information, summarize for me the minimum and maximum price.*

Interviewee: *Summarize results to interviewer and state what you think the minimum and maximum prices should be. Make sure to note that the new pill is a preventative measure, as opposed to the current method of using inhalers as treatment. A good summary will be structured and go back through the steps used in solving the case. At this point you can also bring in other parameters that may not have been discussed in the case such as the price on-patent versus off-patent, or the potential larger international market for this drug.*

Case 8: International Airline

BACKGROUND

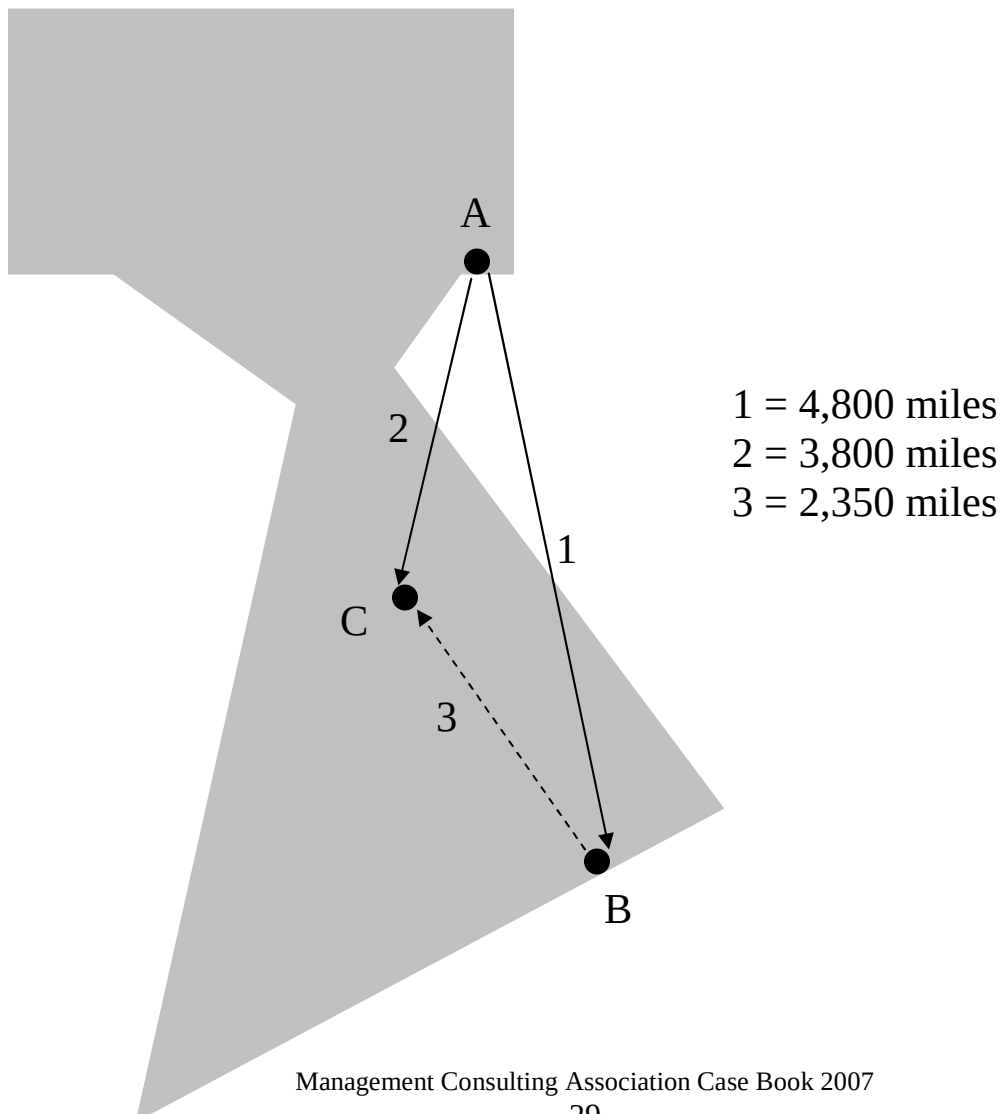
Firm: Bain

Round: 2007 Summer, First

Content: Qualitative and quantitative

CASE QUESTION

Our client is a major international airline with flights to and from the Americas and Europe. The company already operates two daily flights from A to B, offering connections to C. The CEO has hired Bain to evaluate a new route. Should they create this new route from A to C?



INTERVIEWER BRIEFING

Recommended approach: This is a typical Bain case. You can feel free to bring in your understanding of the airline industry as you see fit. This was a real case that the interviewer had worked on during his time at Bain. Take the time to structure your thoughts and discuss the case.

The interviewer was expecting the interviewee to structure a framework, which would look at the big picture that any MBA should have in mind about the airline industry. A strong candidate would have to arrive into final numbers as well as in a final conclusion. Some important elements were:

- First player in the market
- Cannibalization and complementary business
- Discuss trade-offs

Key facts:

- o At this point there is no other airline flying from A to C, and there is only one competitor flying from A to B. B to C is a domestic route with one competitor
- o Market grows with GDP
- o 15% of passengers from A to B (or return) have C as the origin or final destination
- o Current Load Factor in A to B is 80%
- o Ticket price to fly from A to B or A to C is \$600 one way
- o Aircraft type:
 - o Airbus 319 (120 seats) has a maximum fly range of 4,000 miles
 - o Airbus 320 (165 seats) has a maximum fly range of 6,000 miles
 - o Airbus 330 (240 seats) has a maximum fly range of 7,000 miles
- o Costs:
 - o Airbus 319 (120 seats) has a fixed cost of \$41,000
 - o Airbus 320 (165 seats) has a fixed cost of \$62,000

EXAMPLE DIALOGUE

Interviewee: *[Summarize the case and work on a framework]*. This case requires us to estimate the potential market size of route A-C, the growth opportunity, the competition as well as our client's resources. Also, any legal or governmental issue should be discussed.

Interviewer: *OK. How would you estimate the market size (demand) for this route? What do you need to know? I have been working with this client for a long time and might have the needed information.*

Interviewee: Although we could estimate the size of the market, it might be more reasonable to look at the current information the company has about the indirect route. I

would believe many passengers already fly from C to A by connecting flights in B, right? So I would like to find out the number of flights per day, number of passengers per flight and the percent of these passengers that actually have C as the origin or destination.

Interviewer: *Very well thought. It is a smart decision to start more conservative. Our client currently flies twice a day from A to B and back. It currently operates Airbus 320 in this route, with 80% load factor. 15% of the passengers have C as the origin or destination. I can also tell you some aircraft limitations:*

- *Airbus 319 (120 seats) has a maximum fly range of 4,000 miles*
- *Airbus 320 (165 seats) has a maximum fly range of 6,000 miles*
- *Airbus 330 (240 seats) has a maximum fly range of 7,000 miles*

Interviewee: This gives us $165 * 80\% = 132$ per flight or $132 * 2 = 264$ passengers/day (one way), of which 15% or 40 have C as the origin (and destination). We should have in mind that if the client offers the new route, it will reduce the load factor of the A-B route from 80% to 68%. Aside from our own cannibalization, I would expect that by offering this new route our client will be able to take customers from its competitors, right? Actually, does the client have any competition on its current routes?

Interviewer: *There is only one competitor flying from A to B, but this competitor has a code-share agreement with a domestic airline that flies from B to C. From A to B the competitor flies an A330 (240 seats) daily with the same 80% load factor. Also, 15% of its passengers have C as the origin or final destination.*

Interviewee: This gives us an additional $240 * 80\% * 15\% = 30$ passengers/day. Before we conclude that our client will be able to steal these customers from the competition, let's analyze the pricing points. Do you have any information of prices?

Interviewer: *Both our client and the competitor charge \$600/leg from A to B or A to C.*

Interviewee: It is interesting to see that the passenger pays the same fare to fly from A to B (4,800 miles) and from A to C ($4,800 + 2,350 = 7,150$ miles). So basically we have the same price and potentially a faster trip. Unless we find any competitive advantage other than travel time, we could assume our client is able to steal customers from the competitor. I would also imagine that by entering into this route our client would have first-mover advantage and the competitor would not enter this route if they didn't think that it could steal share from our client.

Interviewer: *This makes sense. Basically you have estimated a conservative demand. What else would you analyze?*

Interviewee: We know the potential demand; although there is an expected growth in the industry let's see whether this route would be currently profitable. What do we know about the cost structure?

Interviewer: *The total fixed cost of operating A319 and A320 are \$41,000/per flight and \$62,000/per flight respectively. There are some variable costs, but they are marginal.*

Interviewee: Basically for route A-B the client must use an A320, but A319 is an option for route A-C. Let's analyze each route:

- Revenues from A to B: $(165 * 80\% * 85\%) * 600 = \$67,320$
- Cost for A to B: \$62,000
- Revenues from A to C: $70 * 600 = 42,000$
- Cost for A to C: \$41,000

I am assuming the client could use the A320 for the main route and A319 for the new route. As we can see, the former route will remain profitable (8%) and the new route will give a 2.5% margin.

Interviewer: *What is your conclusion?*

Interviewee: Taking into consideration that 2.5% margin is not a very bad number for the airline industry and that our assumption does not take into consideration an additional demand generated by the new faster route from C to A, the client should at least launch a trial of this new route. The former route, from A to B, will remain profitable.

Case 9: Seattle Newspaper

BACKGROUND

Firm: A.T. Kearney

Round: 2007, Summer

Content: Qualitative and quantitative

CASE QUESTION

Our client is a major media conglomerate, with radio stations, magazines and newspapers throughout the US. They have five major newspapers in five important metropolitan areas. Its oldest newspaper circulates in the Seattle metropolitan area. After 50 years of great profitability and steady market, early this year there was a strong decrease in market share and price.

The CEO has hired A.T. Kearney to help understand the new market dynamics and determine what strategy should be taken next. The CEO is also expecting us to provide advice on whether he should increase prices. How would you go about addressing this question?

INTERVIEWER BRIEFING

Recommended approach: This is a typical case for which one single framework won't be enough. The interviewer was expecting you to look at the big picture and then start thinking about the profitability issue – both cost and revenue – and asking for specific information. A strong candidate would have to arrive at the following set of conclusions:

- The new competitor targets a total different market from our client; this signals that there is an additional market that our client could consider exploring.
- The new sales price is below the cost per paper; however, our client should look at the potential for advertising revenues. These should subsidize the uncovered cost of each paper.

Key facts:

- Market Share: for over 50 years this Seattle newspaper had split the market (50-50) with a strong, but independent local newspaper. The two are well recognized for their news quality.
- Earlier this year, another newspaper entered the market, building a 50% market share in less than nine months and leaving the two former newspapers with 25% market share each. Although the two major newspapers had dropped their prices to match the new entrant's, this has not been enough to recapture the loss in market share. The new

competitor is recognized as providing lower quality news than the former two players.

- Newspapers generate revenues from sales of papers as well as advertisements.
- There is a high fixed cost to operating a news paper, which might be classified as sunk cost.
- None of the players offer subscriptions.

	Before		After		
	Client	Old Competitor	Client	Old Competitor	New Competitor
Price	\$1	\$1	\$0.50	\$0.50	\$0.50
Daily Sales Volume (units)	2,000	2,000	2,500	2,500	5,000
Market Share	50%	50%	25%	25%	50%
Daily Ads (\$)	\$ 2,000	\$ 2,000	\$ 2,000	\$ 2,000	\$ 0
Pages/day (units)	40	40	40	40	20
Cost/page (\$)	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01
Delivery Cost (% of price)	40%	40%	40%	40%	40%

EXAMPLE DIALOGUE

Interviewee: I'd like to understand a few things first in order to evaluate this market, especially what, if anything, has changed from last year. To this end, I'd like to explore the current market demand for newspapers in Seattle. Then, I'd like to understand the competitive landscape. Third, I'd like to discuss any changes to our client's management or internal processes. Finally, since we're dealing with a profitability issue here, I'd like to explore both the revenue and the cost side of the business.

Interviewer: *That was a long list of things. Where should we start?*

Interviewee: You said that there had been a decline in our client's market share, so let's take a look at the competitive landscape. How has the market share of its competitors changed?

Interviewer: *For over 50 years this Seattle newspaper has split the market with a strong but independent local newspaper. The two are well recognized by their news quality.*

Interviewee: Do they still split the market?

Interviewer: *Early this year, another newspaper entered the market, capturing 50% of the market in less than nine months and leaving the two former newspapers with 25% market share each. Why do you think that this happened, what factors are at play?*

Interviewee: The new competitor must be offering either a better product or a lower price point. What do we know about the volumes and prices?

Interviewer: *These are the numbers. What do you think?*

	Before		After		
	Client	Old Competitor	Client	Old Competitor	New Competitor
Price	\$1	\$1	\$0.50	\$0.50	\$0.50
Daily Sales Volume (units)	2,000	2,000	2,500	2,500	5,000

Interviewee: Interesting, actually our client and the former competitor have increased sales volume with the entrance of a new competitor. I would imagine the new competitor is offering a different product or targeting a different market niche.

Interviewer: *That is correct. Although our client has dropped its price to match the new entrant's – and the other major competitor is always following it – this has not been enough to recapture the loss in market share. The new entrant is recognized for offering a lower quality newspaper. So, how would you explain that?*

[Interviewee should have noted the change in price and identified that this may be why there has been an increase in demand]

Interviewee: Well, there are some reasons that I can think of that might explain this phenomenon:

- After the two major newspapers dropped their prices to match the lower price of the new entrant, some of their readers could afford purchasing the two papers.
- Alternatively, potential newspaper readers who could not afford the \$1 newspaper can now purchase newspapers at 50 cents.
- The new competitor captured a new market niche, one that demands lower quality news.

Sales revenue was $2,000 \times 1 = \$2,000$ but now it is $2,500 \times 0.5 = \$1,250$. This represents a 37.5% decline in newspaper sales. I don't think that a newspaper would have such a high profit margin as to cover this decline, let's look at Profits = Revenue – Cost. To do this I need to understand the per newspaper costs.

Interviewer: *Ok.*

	Before		After		
	Client	Old Competitor	Client	Old Competitor	New Competitor
Pages/day (units)	40	40	40	40	20
Cost/page (\$)	\$0.01	\$0.01	\$0.01	\$0.01	\$0.01
Delivery Cost * (% of price)	40%	40%	40%	40%	40%

* There is a cartel for distribution

Interviewee: Our client has $0.01 \times 40 + 40\% \times 0.50 = \0.60 /paper in costs, which is greater than the sales price of 50 cents. The new competitor, on the other hand, has $20 \times 0.01 + 40\% \times 0.50 = \0.40 /paper in costs, with a \$0.10 margin over the variable cost per newspaper. Are there any fixed costs we should be considering or are these considered to be sunk costs? I would expect some heavy costs in this side.

Interviewer: *Yes there are, but they are sunk. However, I can tell you that our client is still making a profit, despite the numbers that you just calculated. How would you explain that?*

Interviewee: Besides paper sales, I would expect that a newspaper generates revenues from advertisements and online. We should also consider the balance between subscribers and newsstand sales. Do you have any information?

Interviewer: *Yes. Neither our client nor its competitors offer subscriptions. They do not have online news either. Let me give you the ad revenues. Any thoughts?*

	Before		After		
	Client	Old Competitor	Client	Old Competitor	New Competitor
Daily Ads (\$)	\$ 2,000	\$ 2,000	\$ 2,000	\$ 2,000	\$ 0

Interviewee: Clearly the new competitor has not impacted the two former players' advertisement revenue. It has also not been able to generate any revenue with ads. How come? I also would want to consider that with the expanded customer base, why there has not been also an increase in advertising, since I would imagine advertisers pay per reader.

Interviewer: *Usually companies split their budget of ads among the different means of communication that reach out to their target customers. So?*

Interviewee: We can conclude that the new competitor has created a new market segment.

Interviewer: *That is correct. So, you have a meeting with the CEO in five minutes, what would you recommend him to do? What will be your strategy?*

Interviewee: I will tell him that he should raise the price, perhaps back to \$1. Some reasons:

- The new competitor is not a direct competitor; it has created a new market niche of people who cannot afford to pay \$1 and reads a totally different kind/quality of news.
- A price of \$0.50 is not feasible in the long run for any of the players due to the high fixed cost.
- As our client already mitigates the high fixed cost, the CEO should think about creating a new product (newspaper) focused on this new market niche. The new product should be totally independent and oriented to this market. In the worst case scenario they would make money out of printing ($5,000 * 0.10$) but could certainly convince its advertisers to direct part of their budgets to the new product.

Case 10: Jumbo Jet

BACKGROUND

Firm: Booz Allen Hamilton
Round: 2007 Summer, First
Content: Qualitative and quantitative

CASE QUESTION

Our client is a large European airplane manufacturer that is developing a new jumbo jet, which will be able to hold up approximately 800-900 people. Parts for the plane are manufactured in multiple sites across Europe. They have provided us the following information on site production:

- Cockpits are produced in Northern Spain
- The fuselage is produced in Northern Germany
- Wings are produced in Northern England
- Landing Gears/Interior are produced in Southern France

The client has asked us to determine where it is most economical to assemble the plane.

INTERVIEWER BRIEFING

Candidate should form a structured approach to investigate the components that make up assembly, including costs and capabilities.

EXAMPLE DIALOGUE

Interviewee: Well, I would like to look into the specific costs incurred from the assembly process and also determine the capabilities for assembling the planes across different locations (labor skill and supply, etc.).

Interviewer: *Ok, I like your approach. Let's focus on costs for now.*

Interviewee: Great. I assume that the major costs of assembly would be transportation, labor, and PP&E. Also, I would think that there would be tax incentives to assembling in one country versus another.

Interviewer: *Good. Let's focus on transportation and labor for now, as these are the most significant cost drivers. What are some of the areas that you might analyze for potential assembly locations?*

Interviewee: I would look at the existing sites where our client manufactures parts, as well as other areas in the region that may have low cost propositions, such as Eastern European countries that may have cheaper labor costs than Western Europe.

Interviewer: *Let's take France as one location since it is the mid-point of the four manufacturing locations and compare it to the Czech Republic where labor is significantly less expensive than Western Europe.*

Interviewee: Well the location in France would incur less cost because the client would already have the landing gears and interior at the facility, and would have to transport the other three parts over a shorter distance. However, the labor costs in France would likely be significantly higher than those in the Czech Republic. Do we have any information about transportation and labor costs in each of these areas?

Interviewer: *Well, we have determined that labor accounts for about 70% of the total costs of assembly and transportation accounts for the other 30%. Disregard the other costs such as real estate, etc.*

Interviewee: Great. Well, since labor accounts for 70% of the cost, let's start there. Do we have any details about how much labor costs in the Czech Republic relative to France?

Interviewer: *Labor in France is 1.5 times higher than labor in the Czech Republic. What do you think that total transportation costs would be in France relative to the Czech Republic?*

Interviewee: Since these parts are so large, I would assume that a major part of transportation is actually packing and loading the pieces on a truck.

Interviewer: *You're right. In fact, the marginal cost of each extra mile transported is close to zero. The most significant cost incurred is labor costs associated with the loading of parts for transportation.*

Interviewee: I would assume that the landing gears and interior are fairly small and wouldn't incur as large expenses for loading/unloading compared to the wings or the fuselage.

Interviewer: *You're right. In fact, the interior and the gears are the easiest to pack of all the four parts and the fuselage is the hardest (most expensive). What would you recommend to our client?*

Interviewee: Well, I would recommend they assemble the plane in the Czech Republic. Since labor accounts for 70% of the total cost, and the labor in Czech Republic is about 35% less expensive than France, the client would save a significant amount of money,

assuming that the skill level is constant across the two locations. Additionally, the transportation costs would not be lower in France since the majority of transportation expense is incurred by loading/unloading the parts and the interior/gears are the easiest to pack. It might be worthwhile to analyze Germany as a potential assembly location because the fuselage represents a significant portion of our transportation costs. The transportation savings may balance out the higher cost of labor in Germany (as compared to Eastern Europe).

Interviewer: *Excellent. One last question – our client knows the demand will be 200 planes per year and they are trying to determine how large an assembly facility they should build. How would you go about figuring this out?*

[Candidate should ask for this data]

- *It takes 8 months to assemble one plane*
- *One plane requires a space of about 300 ft x 300 ft*

Interviewee: Well, we know that it takes .75 of a year to produce one plane, so we need space to produce 150 planes in the facility ($200 \times .75$). Also, one plane requires 90,000 sq. ft. of space. So the total facility would need to be $150 \text{ planes} \times 90,000 \text{ sq. ft. per plane} = 13.5\text{M sq. ft.}$

The interviewee will now be asked to summarize high level findings and make a recommendation to the client regarding where it should produce and assemble planes and the requirements for this operation.

Case 11: Telecom

BACKGROUND

Firm: Booz Allen Hamilton
Round: 2007 Summer, Second
Content: Qualitative and quantitative

CASE QUESTION

Our client is a telecommunications infrastructure manufacturer and they have hired us to help determine if they have any excess manufacturing capacity. If there is any excess capacity, our client would like Booz Allen Hamilton to suggest a course of action to ensure they are profitable and efficient.

The client currently has plants in both Minnesota and Mexico, close to the Texas border. The plants manufacture similar products and the facilities are about the same size. The company expanded capacity by replicating their first plant (using it as a template for others).

INTERVIEWER BRIEFING

Candidate should formulate a logical structure (internal vs. external factors, supply/demand, etc.) and then determine if the client has excess capacity or not. Interviewer does not offer the necessary data and so the interviewee should ask directed questions to extract the following data:

- Each plant operates an 8am – 5pm shift
- Demand is flat or declining
- Client has a relatively high market share of 80%
- Customers are phone companies and there are high switching costs
- Our client recently acquired a competitor and there are very few players in the market.
- The client is not highly concerned about new entrants because the industry growth is flat or declining, this is not an attractive market.

EXAMPLE DIALOGUE

Interviewee: I would start by looking into the company's manufacturing process to determine if they are experiencing low utilization rates compared to the overall capacity of production.

Interviewer: *The company operates 5-8 shifts and is running at fairly close to 100% utilization rate. However, they have considered shifting to multiple shifts in each plant and running a 22-hour workday. What do you think the overall demand capacity looks like in this particular industry?*

Interviewee: I would assume that it is either flat or declining because the US telecom market is relatively saturated, with little new customer growth.

Interviewer: *Yes, that's correct.*

Interviewee: Ok, well, I would now like to investigate the overall market and determine where our client sits in the industry. Do we have any information about their market share?

Interviewer: *They have a relatively high market share of 80%.*

Interviewee: Who are they supplying the products to? Consumers, business customers, or phone companies?

Interviewer: *They are a telecom infrastructure manufacturer so their direct clients are the phone companies.*

Interviewee: Does our client have any reason to believe that the market share could change dramatically? What are the switching costs in this industry?

Interviewer: *There are very high switching costs in the industry. Also, our client recently acquired a competitor so they are not anticipating any big declines in market share. There are also only a few competitors in the industry with little threat of new entrants due to the flat or declining growth.*

Interviewee: Given the overall macro trends in this industry as well as their specific manufacturing set-up, I believe that our client does have excess capacity?

Interviewer: *Why?*

Interviewee: Well, the industry is declining and there is less demand for these products over time. Also, opportunities exist to streamline their manufacturing operations with more shifts per plant across less overall plants.

Interviewer: *Ok. What should they do?*

Interviewee: They could close some existing plants, or they could generate revenues from manufacturing products for competitors (or for companies with similar products).

Interviewer: *Looking at the first option, how would you determine which plants to close?*

Interviewee: I would look at the overall cost per unit at each plant and determine which would result in the most efficient overall location. Costs could be classified into major subsets such as labor, distribution (from Mexico vs. Minnesota). I would also want to know if quality of product differs across any of the plants. Finally, it would be valuable to analyze whether our client could change their plant size and configuration to generate further efficiencies.

Interviewer: *Good. Quality is fairly consistent across all of the plants and they are not looking into plant reconfiguration at this point. Labor costs are the most significant contribution to cost per unit. Given that, what do you recommend?*

Interviewee: I recommend our client closes the Minnesota plants and increases overall utilization of the plants in Mexico by adding shifts.

Interviewer: *What are the major risks/concerns of this recommendation?*

Interviewee:

- Negative PR resulting from closing US plants
- Scalability issues (can the Mexican plants quickly increase capacity?)
- Quality issues stemming from increased production
- Reliance on one location/region for all manufacturing

Case 12: Mining in Brazil

BACKGROUND

Firm: BCG

Round: 2007 Summer, Second

Content: Qualitative and quantitative

CASE QUESTION

Our client is a US industrial conglomerate, with major investments in South America, India and China. One of these investments is a mining operation in Brazil. At this mining operation, our client produces only one metal, which is considered to be an international commodity product. This metal has hundreds of applications. In Brazil there are only two other producers.

The CEO has hired BCG to help identify new opportunities for this business as well as understand the market dynamics. He wants to know whether he should divest the mining business or invest in an additional facility. This afternoon, the BCG team is going to meet with the CEO to discuss our initial hypotheses. We have been provided the following information to assist us:

Local Players	Plant capacity (tons)	Cost/ton	Sales	
			Local	Export
Client (A)	600,000	\$450	600,000	-
Competitor B	4,500,000	\$420	2,800,000	1,200,000
Competitor C	3,200,000	\$420	2,000,000	700,000

The world-wide production (supply) of this product is estimated to be 100,000,000 tons. Furthermore, it is known that demand for this product is greater than current supply. As a commodity, the international price (export price) is \$450/ton and the local price is \$600/ton.

INTERVIEWER BRIEFING

Recommended approach: This is a typical BCG case. BCG often (i) uses real life cases, (ii) presents available data upfront in a table of during the case question, and (iii) drives the interview to be a conversation between the interviewer and the interviewee. Therefore, be sure to engage the interviewer as you approach the case.

The interviewer was expecting the interviewee to first provide a structure/framework, which would look at the big picture and then start hypothesizing. A strong candidate would:

- Discuss market dynamics (local and international and supply/demand).
- Discuss the expected competitive response to any action, e.g., a price war.
- Summarize all findings in a presentation format.

Key facts: All given beforehand, but the following would help in the discussion:

- o An efficient plant should have a 1,000,000 ton capacity (but, not all plants are operating efficiently), however, from this information the interviewee should be able to assume that therefore competitors are operating more than one plant each.
- o Market grows with GDP.
- o There is a strong demand for this product internationally.
- o The competitors are probably located away from the coast, adding transportation costs.

EXAMPLE DIALOGUE

Interviewee: *[Summarize the case and work on a framework]*. In this case it is important to look at the competition (specifically, understand the different cost structure of the 3 producers), estimate the market demand and discuss the international trade environment. We should also discuss the specifics of a metal commercialization.

Interviewer: *OK. Where should we start?*

Interviewee: Clearly, our client is running on full capacity, but its competitors appear to have some idle capacity. Furthermore, I note from the data provided that the competitors' cost structures allow them to sell in the international market while our client does not currently export any of the products. Our client would experience zero margin if it was to export since the international price is given at \$450 which is equal to our client's cost of production. Nevertheless, the local price is considerably higher than the international price, so the producers would rather sell as much as possible in the local market. Given this information, I would hypothesize that there is not enough demand in the local market.

- o Competitors would prefer to sell in the local market (\$600/ton) instead of export (\$450/ton)
- o Competitors are running with idle capacity, but we know that demand exceeds current supply

Interviewer: *That is correct. But what about the international market?*

Interviewee: This is a good point. You did mention that there is a strong demand worldwide. Now we have to find out why the competitors are not selling their full capacity. We can think of many possible reasons. Geographical distance, transaction costs, transportation costs, export taxes, etc.

Interviewer: *They all make sense, but as we see competitors are able to export some metal, right?*

Interviewee: Competitors might have operations abroad so it makes easier to export to their international facilities. Or they might produce part of their capacity close to harbors. Considering that the international price is much lower than the local one, I would expect some barriers for international trade.

Interviewer: *That is correct, Brazil does have some taxes for foreign products and producers struggle with transaction costs. Let's look at the cost structure. Why do you think there is a difference in costs?*

Interviewee: I could consider geographic location, technology, economies of scale, supply chain synergies, etc.

Interviewer: *You mentioned some important elements. Our client has only one location and the competitors have multiple plants, each in the countryside. Our client was the first player in this market, having built its facilities 10 years earlier than its competitors. The industry has realized that a plant must have a minimum of 1,000,000 capacity in order to be efficient.*

Interviewee: But our client is still making a lot of money, right?

Interviewer: *Yes, enough to convince the CEO to invest in a new production facility. This would be a \$400M investment in year 0 for a capacity of 1,000,000 tons with a cost of \$420/ton. How would you evaluate this investment if this new production would be traded in the international market? Would you recommend this investment?*

Interviewee: Basically the margin will be \$30/ton $(\$450 - \$420) * 1,000,000$ equal to \$30M per year. By using a 10% discount rate this investment will generate \$300M in total. The \$400M investment would not be worth it.

Interviewer: *That is true. So, are you going to advise him not to make the investment?*

Interviewee: I would advise him against it unless he is willing to engage in a price war.

Interviewer: *That would be the minimum price he could go to turn this investment profitable?*

Interviewee: We should be cautious because a lower price would impact the current profitability. The current margin is $(\$600 - \$450) * 600,000 = \$90\text{M}$. For instance, if we drop the price to \$513/ton we would be losing \$52.5M $[(\$600 - \$513) * 600,000]$ from the current 600,000 ton production but generating \$93M $[(\$513 - \$420) * 1,000,000]$ from the new production, a net gain of \$40.5M. With the 10% discount the minimum profit required from the new plant would be \$40M $(\$400\text{M} * 10\%)$. Therefore, the minimum price would be \$513.

Interviewer: *But if you drop the price to \$513/ton, what would the competitors' reaction be? Would they also increase production? Remember, they have idle capacity and you don't.*

Interviewee: That is correct, but the competitor has a lot more to lose with a price reduction. In our client's case we found out that it would lose money as the margin of the current production drops. However, the client only sells 600,000 tons right now while the competitors sell 4,800,000 tons combined. They would probably reduce their production to avoid a higher price reduction.

Interviewer: *Really? So you are recommending our client to invest \$800M in a 2,000,000 tons capacity plant?*

Interviewee: I haven't done the math but I guess this would be too risky. I would recommend our client to invest the \$400M and see how the market reacts.

Interviewer: *That is a fair recommendation; after all they will be playing a game with no real expected result.*

The interviewee should now expect to summarize the findings from this discussion for the client, highlighting the approach and key recommendations.

**THIS IS THE LAST PAGE
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**COLUMBIA
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Management Consulting Association

Case Book
2006



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Case 1: Bank Increasing Market Share

BACKGROUND

Firm: N/A

Round: 2005 Summer, First

Content: Mainly qualitative

CASE QUESTION

Our client is a major diversified financial services firm. They have 4 lines of business:

- Checking Accounts
- Savings Accounts
- Loans
- Mortgages

The bank is profitable, but the CEO is on a mission to increase overall market share while still maintaining profitability. He is considering increasing his sales force to accomplish this.

There are 3 different categories of sales force he can choose to change:

- Branch Managers
- Telemarketers
- Mortgage Agents

The CEO has come to us for advice on whether or not he should increase his sales staff, and if so, where he should be adding heads. How would you go about addressing this question?

INTERVIEWER BRIEFING

Recommended approach:

A thoughtful and complete answer to this case involves discussing the overall market environment, with a look at the competition in the marketplace and the current market share breakdown. The interviewee should realize that there are 4 separate markets for distinct products that have different profitability characteristics. It becomes apparent that investing in some lines of business will pay off for the company more than investing in others. From there, a discussion of the economics of each of the different sales channels will drive you towards a recommendation.

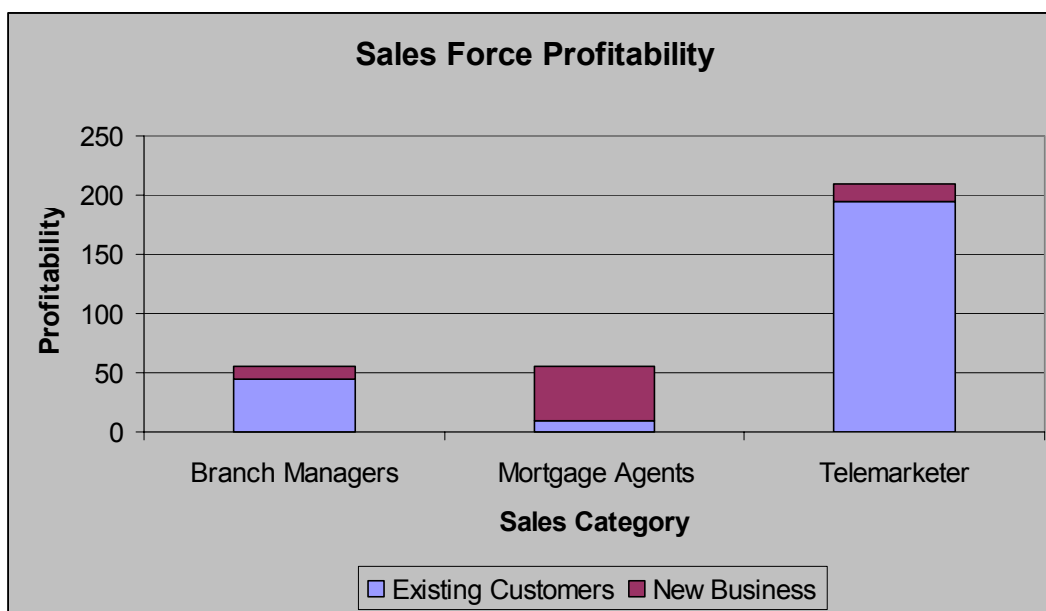
1. Determine the competitive position of the company
 - What is the competitive position of the company in the overall market?
2. Assess the importance of the different lines of business
 - Where does the bank play? What is its strategy? How does it make its money?



- What is our market share in each line?
 - What is the importance of each line to the overall business? How do the different lines fit into the overall strategy? Which are the most profitable? Are there any loss leaders?
 - What can the company expect to achieve from investment in each line?
3. Understand the impact of the different categories of sales force
 - Which lines of business will be affected by increased investment in each category?
 - What is the expected ROI for each category?
 4. Formulate an investment strategy for the bank to follow

Key facts:

- The market is made up of 5 major players. They are all very similar, offering diversified services and playing in all 4 segments.
- Market share is defined on a % of customers basis. The market shares are as follows – the #1 player has 20% market share. We are the #2 player and have an 18% market share. The rest of the industry is somewhere between 15%-18%.
- The market share within each of the segments does not break down in this same fashion. Everyone is profitable and established and no one is pricing their products irrationally.
- The company's share by segment is as follows:
 - Checking Accounts – 21%
 - Business Accounts – 20%
 - Loans – 23%
 - Mortgages – 10%
- Profit per transaction is as follows:
 - Checking Accounts - \$5
 - Business Accounts - \$20
 - Loans - \$100
 - Mortgages - \$1,000



Editor's note: The chart above depicts the profitability of different types of sales agents with regard to the mortgage business. "Profitability" of each individual sales agent is defined as the incremental \$ profit margin (in \$K) each brings in. Telemarketers clearly bring in the most profit margin \$s in aggregate, but mortgage agents bring in the most from new business.

EXAMPLE DIALOGUE

Interviewee: Before we start, I would like to clarify what a mortgage agent actually does. Can you walk me through that?

Interviewer: *The mortgage agents work by having relationships with real estate brokers. When a broker makes a sale, he refers the buyer of the home to a mortgage agent. That agent works with the home buyer to close the transaction and sell the mortgage.*

Interviewee: That makes sense. I'd like to figure out what the current market situation looks like. How many players are there and what is the breakdown in terms of market share? Is everyone profitable? Are there any new players? Any new trends like internet banking?

Interviewer: *The market is made up of 5 major players. They are all very similar, offering diversified services and playing in all 4 segments. The market shares are as follows – the #1 player has 20% market share. We are the #2 player and have an 18% market share. The rest of the industry is somewhere between 15%-18%. However, the market share within each of the segments does not break down in this same fashion. Everyone is profitable and established and no one is pricing their products irrationally. There are some new trends emerging such as internet banking, but they are not a factor at this point, and the CEO is not worried about that.*

Interviewee: I see. Do we know how our market share breaks down in each of the different segments?



Interviewer: *Yes, we do. Our breakdown by segment is as follows:*

- *Checking Accounts – 21%*
- *Business Accounts – 20%*
- *Loans – 23%*
- *Mortgages – 10%*

Interviewee: That is interesting. Is there a different profitability level for each of the different segments?

Interviewer: *Yes, there is. As a matter of fact, profitability per transaction is as follows:*

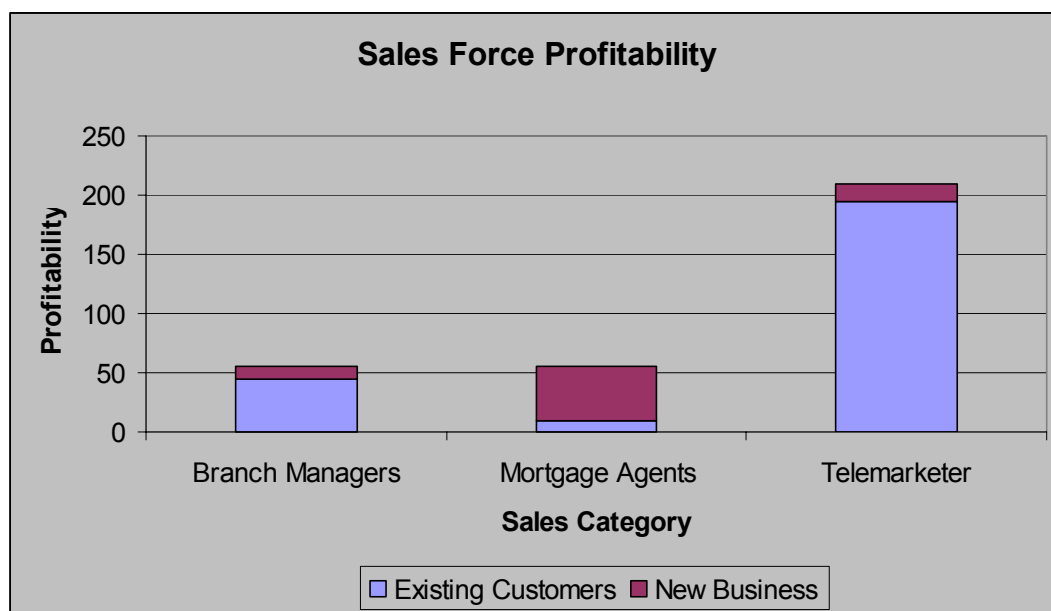
- *Checking Accounts - \$5*
- *Business Accounts - \$20*
- *Loans - \$100*
- *Mortgages - \$1,000*

Interviewee: I would like to explore a few more possibilities, but first, I want to just say that my gut instinct is that we can increase our sales force, and I think that we should do it in the mortgage agent area.

Interviewer: *That is a good insight. Why do you think that is?*

Interviewee: Based on the existing market shares, it seems like we have the most room to improve our business in this area. Additionally, given the parity between the competitors in the industry overall, I think it would be very difficult to dramatically increase share in any of the other areas. Therefore, we should concentrate on the mortgage market. From there, we need to see what part of the sales force will have the greatest impact on this market. Do we have any information on the profitability of each of the salespeople and how much new business each arm brings in?

Interviewer: *Yes, the CEO has given us this exhibit:*



Interviewer: *What does this chart tell you?*

Interviewee: This shows that telemarketers are the most profitable place to put our money in terms of increasing our sales force. How are they generating these profits? What are they actually doing?

Interviewer: *They call up existing customers to get them to renew their old services.*

Interviewee: I see. In that case, it looks like my initial hunch was correct. Since the other 2 sales forces will increase profits based on existing customers, the mortgage agent is where we can drive market share growth by winning new customers. In terms of making an overall profit, the telemarketers would be the right way to go. However, since the CEO has explicitly stated that he wants to increase market share, mortgage agents are where he should be bulking up the sales force.

Interviewer: *Thank you. Out of curiosity, can you give me an example of where adding branch managers might be a good decision? What would you need to look at?*

Interviewee: We could do a study of each branch to see how many customers come through the door and how many of them are existing customers versus potential new customers.

Interviewer: *OK. Let's say we did that and we found out that in each store, there are 4 branch managers and they, on average, service 40 potential new customers that come through the door everyday. What would you have to look at when deciding whether or not to add more branch managers?*



Interviewee: I think you would have to look at whether or not the existing 4 managers can handle 10 new customers per day. If they can handle that amount of customers, then there is no need for more managers. In that situation, if we were to add a 5th manager, each manager would then be responsible for 8 new customers on average, but that would just create idle time and not generate any new customers. If however, there are long waiting times and potential new customers are getting fed up and leaving, then perhaps adding new managers would make sense.

Interviewer: *What else could change that could cause the bank to want to add more branch managers?*

Interviewee: If you can drive additional potential customers through the door. Perhaps a changing trend in the market will create a new opportunity, i.e. adding a new product that may draw in more non-customers who are interested in becoming a customer or a new advertising campaign that is drawing in more people. In any of those types of scenarios, it may make sense to add additional branch managers.

Interviewer: *I think that makes a lot of sense. Thank you very much.*

Case 2: Chinese Cars

BACKGROUND

Firm: Roland Berger

Round: 2005 Summer, First

Content: Market sizing, qualitative and quantitative

CASE QUESTION

Our client is a large Chinese conglomerate with a lot of cash on its hands. They decided to invest this money in a plant that, by the end of this year, will have a full production capacity of 8 million aluminum wheels annually. Management assumes that it will be able to sell 20% of its production in the domestic Chinese market.

The company hired Roland Berger to determine if it can sell the remaining 80% of its production to the US market, which is the largest vehicle market in the world. What do you think?

INTERVIEWER BRIEFING

Recommended approach:

This case is essentially asking you to do two things:

- 1) Determine the size of the US car market so that you can understand the relative impact of the Chinese company's goals.
- 2) Assess the feasibility of entering the US market and provide suggestions on how this company might go about doing so.

Market Sizing: Determine how many wheels your client could supply to the US market – both in number of wheels and number of cars (1/4 of the wheel number). Then ask about the size of the US new car market and the secondary market to determine what share the company would need to capture to sell 80% of its production in the US.

Feasibility and Possible Actions: This is where you need to drill down to uncover automobile industry dynamics:

- How do OEMs get the wheels for the cars they produce?
- How would a Chinese company distribute its wheels to the US market? Directly or through intermediaries?
- How entrenched are supplier relationships? Do OEMs buy based on price only, or do the relationships matter too?
- What regulations are involved in supplying an OEM?

Once you understand how the supplier-OEM relationships work, you can provide some suggestions on how this company might feasibly enter the US market.

Key Facts:



- 17 million vehicles produced in the US each year. 70% equipped with aluminum wheels.
- 10% of used car owners whose cars are less than 5 years old buy aluminum wheels
- 85 million cars in the “after market,” 24 million of which are less than 5 years old
- Most companies will select one supplier who will supply the aluminum wheels for the entire life-time of a car model. Suppliers are usually selected 24 months before a new model goes into production and have stringent quality controls.

EXAMPLE DIALOGUE

Interviewee: Okay. I’d like to first determine the production figures of the Chinese manufacturer and compare them to the size of the total US automobile market. This will help me understand the relative size of the Chinese company’s production numbers and whether their plan is realistic. Once I know how much market share the Chinese company wants to capture, I’d like to discuss the US automobile industry dynamics to determine if/how the Chinese company should enter the market.

Let’s start with their production numbers. Let’s assume that cars require only 4 aluminum wheels and that spare tires are made out of some less expensive material. This means that your client produces wheels for 2 million cars per year. 80% of 2 million gives us wheels for 1.6 million cars, which we are trying to sell in the US. So how big is the US new car market and how large is the secondary market (people who buy aluminum wheels for an older car)?

Interviewer: *About 17 million vehicles are produced in the US each year. About 70% are equipped with aluminum wheels. About 10% of used car owners whose cars are less than 5 years old buy aluminum wheels.*

Interviewee: That means that roughly 12 million new cars each year are sold with aluminum wheels. Can we assume that there are about 280 million cars in the after-market, about 1 car per person, considering that people in New Jersey own on average 2 cars?

Interviewer: *No, that number is much lower. There are about 85 million cars in the after market, but only about 24 million are less than 5 years old.*

Interviewee: I see. That means the after-market is about 2.4 million sets of aluminum wheels per year. Considering that your client needs to sell 1.6 million sets, they’d have to capture a 66% market share, which seems very unlikely in such a fragmented market. That means we should probably focus on the new car market. How do the major car companies procure the aluminum wheels they put on their cars?

Interviewer: *That’s a good question. Most companies will select one supplier who will supply the aluminum wheels for the entire life-time of a car model. Suppliers are usually selected 24 months before a new model goes into production and have stringent quality controls.*

Interviewee: We probably would have to add another 12 months to even get on a company’s preferred supplier list, which increases the lead time to about 3 years before we can hope to



supply aluminum wheels to a major US car manufacturer. Since you mentioned earlier that your client's plant will be ready for production by the end of this year, this does not look like a viable option either. What about trying to expand distribution on a broader basis than just the US?

Interviewer: *They thought about that as another option, but they would really like to work with just the US.*

Interviewee: Assuming that your client can produce quite a bit cheaper in China than most competitors can in the US, it might make sense to sub-contract some of this production capacity to suppliers of the major US car manufacturers. This is probably less lucrative than selling directly to end-customers or car manufacturers because the client would have to share its profit margin with the supplier. That said, it would probably be the most efficient way to enter the US market on a large scale.

Interviewer: *Yes, this is exactly what they did.*

Case 3: Health Insurance

BACKGROUND

Firm: Booz Allen Hamilton

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE QUESTION

A health insurance company experienced a loss of \$40M this year, after recording a \$40M profit last year. They have hired us to determine the cause(s) of this decline and identify potential solutions.

The company is in only one line of business, Commercial Risk Insurance, in only one region. They have several direct competitors.

Their cash flows originate from premiums, which are paid in on a periodic (e.g. monthly, semi-annually, annually) basis by its customers, and claims are paid out to the customers' caregivers as appropriate.

For the sake of simplicity, the company's Profit = Premiums – Claims.

Currently this company has 4 million members, which equates to a 40% market share.

All revenue, cost and profit figures are measured on a per-member basis.

How would you go about identifying potential solutions for their decline in profitability?

INTERVIEWER BRIEFING

Recommended approach: This is a typical profitability case, with the addition of a problem-solving section to discuss the area that is hurting profitability. First, you should hone in on what is causing the company to lose money. Look at revenues (price X quantity) and costs (fixed + variable costs). This will lead you to the crux of the problem. At that point, you should ask questions that explore the problem and identify ways to address it.

Key facts:

- Revenues have been stable from last year to this year. Premiums charged per member have been stable at an average of \$200/year, and membership has been flat at 4 million members.
- The company has made no large recent capital expenditures
- Per member variable costs are rising by \$20/year. This is largely the result of a decline in utilization of resources due to poor management
- Within the company's structure, there are several functional groups:



- New Admits
- Pre-authorizations
- Case management
- These groups are staffed with MDs and RNs that used to be in the field treating patients. They are now responsible for managing hundreds of doctors and nurses that are currently treating patients. They are more used to working in practices of 5-10 professionals and unaccustomed to managing such a large staff, causing problems in controlling referrals and unnecessary testing/procedures.

EXAMPLE DIALOGUE

Interviewee: Since we're dealing with a profitability issue here, I'd like to explore both the revenue and the cost side of the business. Starting with the revenue side, have the premiums charged changed over the last year?

Interviewer: *No, they have been stable. Incidentally, premiums are charged at an average of \$200/year/member.*

Interviewee: Has the member base been stable? The reason I'm asking this is to understand whether revenue as a whole has been stable.

Interviewer: *The member base has been stable, so revenues have as well.*

Interviewee: OK. Since revenue is stable, I'd like to explore the cost side of the business. First, let's take fixed costs. Have they made any large capital investments to spur growth?

Interviewer: *No. They aren't in a growth stage and haven't made any large capital expenditures except for a new claims payment system two years ago. In general, this is not a capital intensive business.*

Interviewee: That makes sense since it is a service company. So let's look at variable costs. Has the cost of servicing members been growing?

Interviewer: *Yes. Per member costs are rising to the tune of \$20/member/year. Why do you think this is happening?*

Interviewee: The cost of care may be going up – on the doctor, equipment or medicine front.

Interviewer: *The biggest cost driver is actually utilization, i.e. how frequently the doctors use other sources for a patient's care. So now I'd like you to derive the total variable costs that the company had last year versus this year.*

Interviewee:

Profit = Revenue – Costs

\$40M (last year) = (4M members * \$200/year/member) – Costs



$\$40M = \$800M - \text{Costs}$
 $\text{Costs} = \$760M \text{ last year}$

$\text{Profit} = \text{Revenue} - \text{Costs}$
 $-\$40M \text{ (current year)} = (4M \text{ members} * \$200/\text{year/member}) - \text{Costs}$
 $-\$40M = \$800M - \text{Costs}$
 $\text{Costs} = \$840M \text{ current year}$

Interviewer: *What were the variable costs on a per member basis for each year?*

Interviewee:

$\text{Per Member Variable Costs} = \$760M \text{ last year} / 4M \text{ members} = \$190/\text{member/year}$
 $\text{Per Member Variable Costs} = \$840M \text{ current year} / 4M \text{ members} = \$210/\text{member/year}$

Interviewer: *Good. Let's talk about utilization now. What might affect this measure?*

Interviewee: It could be a change in what MDs want to prescribe. For instance, maybe they are more risk-averse because of a rise in malpractice cases, so they are more frequently seeking second opinions. Another thing that might affect utilization is how well the company manages the MDs.

Interviewer: *To your second point, what would you look for within the company to see where the problem resides?*

Interviewee: I would look at several things, including strategy, skills of the MDs, shared values, staffing, technology used, leadership system and functional organizational structure, to name a few.

Interviewer: *OK. What would you want to know about the organizational structure?*

Interviewee: I'd like to know how the company is structured.

Interviewer: *Within the company's structure, there are several functional groups:*

- *New Admits*
- *Pre-authorizations*
- *Case management*

These groups are staffed with MDs and RNs that used to be in the field treating patients. They are now responsible for managing hundreds of doctors and nurses that are currently treating patients.

Interviewee: Are those MDs and RNs used to managing large groups of people (i.e., **many** doctors in the field)?

Interviewer: *No, they are more accustomed to working in practices of 5-10 professionals.*



Interviewee: Makes sense. What types of people compose the upper management of the company?

Interviewer: *They are also MDs that used to be in the field. What's your hypothesis about this problem and potential solutions at this stage?*

Interviewee: The problem seems to be in the management of the doctors and nurses that are treating the patients. Due to the difficulty in getting all necessary data, and the fact that the staff managing those doctors/nurses is not used to managing a large, remote organization, variable costs (specifically the costs associated with utilization) have risen. I would recommend reviewing the current information systems for a way to consolidate the view of data for each job function so that everyone has all the necessary information to manage their cases. They should also consider providing leadership/management training for those who are responsible for case management.

Case 4: Italian Paper

BACKGROUND

Firm: Bain

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE QUESTION

Our client is an Italian paper manufacturer. They make high-end glossy and matte paper that's used in corporate brochures, magazine ads for Gucci and Prada, etc. It's a commodity product.

They have one factory that produces 150,000 tons of paper a year. 90% of this is sold into the Italian market, which gives it about a 20% market share. The other 10% is sold outside of Italy.

They would like to build another factory to produce another 150,000 tons a year. This would cost \$100 million. They would like us to tell them whether or not this is a good idea.

INTERVIEWER BRIEFING

Recommended approach: This case is asking you to determine whether it makes sense to build additional capacity, i.e. can your company be profitable in the future based on this investment? This will depend on the current and future market dynamics, which can be explored in a number of ways.

Having the Three Cs in your head will help you ask the right questions:

- **Customers:** Who are they? Is there enough market demand for additional capacity?
- **Competitors:** Who are they? Where are they? What market share do they have? How are their prices relative to ours? Can we capture any of their share?
- **Company:** How much will this additional expansion cost the company, and what will the benefits be? Does this fit with the company's current plans?

Knowing the answers to these questions will then help you determine whether it makes sense to add capacity. You can also probe whether a new factory is the right way to go, or whether it might be more feasible and profitable to add capacity to an existing plant, acquire another company, etc.

Key facts:

- The Italian market will continue to grow at about 10%. Germany is very hot and growing at about 15%. Other Eastern European markets are growing as well.
- Your largest Italian competitor has 40% share. An international competitor has 10% and the other 30% is fragmented among small players.



- There are about 12 players and only 2 paper machine manufacturers in Europe. Everyone knows what everyone else is doing and nobody else is investing in additional capacity right now.

EXAMPLE DIALOGUE

Interviewee: I'd like to understand a few things to evaluate this decision. First, I'd like to explore the current market demand for paper. Then, I'd like to understand the current players in the market and how our company competes against them. Finally, I'd like to talk about the specifics of the investment proposed and what the NPV might be for that option versus others.

I'll start with the market demand in Italy. You mentioned that 135,000 tons is about 20% of the Italian market, which means that the Italian market is 675,000 tons. What are the projected trends for Italian market growth?

Interviewer: *Well, how would you go about finding that out?*

Interviewee: I might look at historical trends, but those are probably not that accurate because of the growth of the Internet. To get a more accurate picture of where the industry is going, I would interview the client's customers, i.e. publishers, printers and ad agencies. I would ask them how they see their business prospects, what they are investing in now, what they want to invest in later, etc.

Interviewer: *OK. Through this, you discover that the Italian market will continue to grow at about 10%. Your largest Italian competitor has 40% share. An international competitor has 10% and the other 30% is fragmented among small players.*

Interviewee: Interesting. So that means the Italian market could only absorb an additional 67,000 tons - not an additional 150,000 tons. It might make sense to look beyond Italy depending on the industry dynamics. What is the greater European market demand?

Interviewer: *Well, Germany is very hot and growing at about 15%. Other Eastern European markets are growing as well.*

Interviewee: Great. Who are the major competitors in those markets and what kind of share do they have?

Interviewer: *There are about 12 players and only 2 paper machine manufacturers in Europe. Everyone knows what everyone else is doing and nobody else is investing in additional capacity right now.*

Interviewee: Are they not investing because they don't see opportunity, or because they don't have the resources?

Interviewer: *It's not clear.*



Interviewee: Turning to the investment angle, how quickly can we make back the money on the investment in the new plant?

Interviewer: *I don't know the specifics, but it wouldn't take very long. What would you advise, knowing what you know now?*

Interviewee: While I don't have all the information I would like, it seems that the European market could absorb the demand and since the competitors are not investing in additional capacity, this could be a great opportunity for the client to make inroads into a new market and increase revenues. Besides the new factory, they may also want to consider other growth options, such as building new capacity in an existing plant or acquiring a competitor.



Case 5: Sugar Magnolia Hospital

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE QUESTION

Your client is the CEO of the Sugar Magnolia Hospital. The hospital is a large hospital providing a full range of services, in a large, urban area. In the last five years, the hospital's profitability has decreased to the point that they are almost out of money and will not be able to meet their financial and social mission.

The revenue scheme of the hospital's different services falls into the following three categories:

- 1) Fixed fee for service (e.g. broken leg = \$150 to fix)
- 2) Cost plus (cost of providing the product/service plus a percentage)
- 3) Per diem fees (fee per day of hospital stay)

The hospital gains patients through physician referral. In other words, the physicians within the hospital see patients and after diagnosing those patients will refer the patients for care, treatment, surgery, testing, etc. to other places within the hospital.

The CEO comes to you to ask the following:

- 1) Why has profitability gone down?
- 2) How should they turn it around?

INTERVIEWER BRIEFING

Recommended approach:

This is a profitability problem, so plan to drill down on Revenues (price and quantity of patients or patients/ services at the hospital) and Costs (fixed and variable costs per patient.)

Ask about the economics of the case:

- How does the hospital make money? (Paid in the revenues structures given by HMO, insurers, and state agencies above. This is given upfront in the case description and is a bit of a red herring in the case.)
- Who pays the costs? (Again, the costs are paid by the HMOs, insurers, and state agencies)
- How do patients come into the hospital? (Patients come in through referral from the doctors that work in the hospital or through emergency services. This is helpful information to get in solving what is going on with revenues in the case. When one segments the products/ services in the hospital, one should think about patient/ service segments, or patient/ doctor segments.)



After receiving this information, focus in on the following:

- How many more people would you be able to see if you reduced patient stays on average by 1 day?
- What would the effect on costs be for reducing patient/ days by one day on average? How would that affect profitability?

Key facts:

There are several patient/ service segments in the hospital. The following are a few examples:

- 1) Women in Labor/ Giving Birth- Low profitability, High referral rates/ word of mouth, volume declining, fixed fee rate with high variability in length of stay
- 2) Pediatrics- Low profitability, high referral rates/ word of mouth, volume declining, fixed fee rates
- 3) Males (age 25-50)-Medium profitability, medium referral rates, volume stable, Per diem and cost+ rates
- 4) Elderly (65+)-High profitability, low referral rates, volume increasing, fixed fee rates

Quantitative work in terms of cost:

Fixed costs: \$1000/ patient

Variable costs: \$500/ patient/ day

Average length of patient stay: 10 days

Patients seen per year: 1000

EXAMPLE DIALOGUE

Interviewee: This is a profitability case, and profitability is composed of revenues and costs. In terms of which side of the hospital's profitability I'd like to start with, unless costs have changed in some significant way because of a recent capital expenditure....?

Interviewer: *No, they've had no new costs.*

Interviewee: Right, then I'd like to start with revenues and see if there have been any recent changes on the revenue side.

Interviewer: *Sure, that sounds like a good place to start.*

Interviewee: So, revenues are composed of two things: price times quantity. Have the prices that the hospital charges for services that it provides changed recently?



Interviewer: *No, prices haven't increased or decreased in any considerable way—other than general increases along with inflation.*

Interviewee: Okay, so, it's a problem with the number of patients, or number of patient/ services, along different segments then.

Interviewer: *Yes, you're right. What do you hypothesize is happening?*

Interviewee: I imagine that there are several patient, or patient/ service, segments within the hospital.

Interviewer: *That's right.*

Interviewee: Is the hospital a full-service provider, or do they specialize in some sort of area, or service?

Interviewer: *Nope, the hospital is a full-service provider.*

Interviewee: So, I hypothesize that more profitable and/ or higher revenue generating patient/ service segments are decreasing.

Interviewer: *Interesting. There are 4 principal segments:*

1) Women in Labor/ Giving Birth- Low profitability, High referral rates/ word of mouth, volume declining, fixed fee rate with high variability in length of stay

2) Pediatrics- Low profitability, high referral rates/ word of mouth, volume declining, fixed fee rates

3) Males (age 25-50)-Medium profitability, medium referral rates, volume stable, Per diem and cost+ rates

4) Elderly (65+)-High profitability, low referral rates, volume increasing, fixed fee rates

What do you think this data tells you?

Interviewee: Well, first, I would prioritize the segments in terms of how important they seem to our current inquiry: I would put the third segment aside because the volume has stayed stable, and I would concentrate on the first two segments for two reasons: 1) Their volume has been declining, and 2) These segments reflect high word of mouth and long-term patient segments that are likely to keep coming back to the hospital for other services for a long time if they like the service that they get there (as opposed to the elderly segment, which represents a diminishing group of patient/ services because of their age).



Interviewer: *Good, good. And what would you want to do in order to reverse the decline of those segments?*

Interviewee: I would probably want to do some research into why those segments' volume is declining. Has service quality declined? Have the number or quality of pediatricians or obstetricians at the hospital declined? Has another hospital in the area begun to specialize in pediatrics and obstetrics that is making this patient and patient/service volume decrease in these specialties?

Interviewer: *Good, good. One last question: would you abandon the fourth segment (elderly) patients completely. They are a highly profitable segment. How would you deal with them?*

Interviewee: They are a highly profitable segment, which is a bit strange, as they have a fixed fee rate revenue structure and their length of stay can vary tremendously. I would try to segment them further into patient/ service segments that were a little more predictable along the dimension of how long they stay—this might yield a much more profitable and predictable patient/ service type than what they currently have.

Interviewer: *Right, so there might be some 20% of those patients that are staying longer and driving down profitability for the rest?*

Interviewee: Yes, and a more detailed segmentation might show that. So they'd be able to target and get the more profitable of those patients.

Interviewer: *Good, good. So, that pretty much covers the revenues side. Now, suppose that I told you that you told you that we went to the CEO with all of this interesting info about revenues and revenue generation by targeting different segments, but the CEO said that they really thought it was about the number of patients that the hospital can see and the costs and cost controls. How would you think about costs?*

Interviewee: Well, there are two components to costs: fixed costs and variable costs. Do we know fixed costs and variable costs by patient, or by patient/ service type?

Interviewer: *Well, actually, we don't have cost data by patient or by patient/ service type, like we did in the revenues discussion. But we do have cost data per patient on average at the hospital.*

Interviewee: So, aggregate data on a per patient level?

Interviewer: *Yes. We know that on average the fixed cost for admitting a patient is \$1000. And the cost per patient day is \$500. The average length of a patient's stay is 10 days. And the hospital sees 1000 patients each year. The CEO is curious about how many more patients the hospital could see if they reduced the length of an average patient day by one day.*



Interviewee: Well, currently they are seeing 1000 patients annually. Each patient stays 10 days on average; so, the hospital has the capacity for 10,000 patient days annually (1000 patients X 10 days = 10,000 patient days). If you reduced the stay to 9 days on average for each patient, then there would be an excess of 1000 patient-days of capacity (10,000 patient-days minus 9000 patient-days). If each patient stays 9 days on average, then the hospital could see about 111 more patients each year (1000 excess patient-days / 9 days = 111 patients).

Interviewer: *That's right. Now, how much would costs decrease if they did this?*

Interviewee: Well, currently, they see 1000 patients per year for an average of 10 days each. Fixed costs for these patients are \$1000 X 1000 patients, which equals \$1,000,000. Variable costs are 10,000 patient days X \$500/ day, which equals \$5,000,000. So, their current costs are \$6,000,000.

If they reduced the average patient stay by 1 day, then their variable costs change. They still have the same fixed costs of \$1,000,000. But their variable costs have changed: they now have 9,000 patient days X \$500/ day, which equals \$4,500,000 in variable costs. That's a cost savings of \$500,000 each year.

Interviewer: *That's right. So, how would you summarize these findings to the CEO and what recommendations would you make?*

Interviewee: On the revenues side, Sugar Magnolia is losing patients in a few segments that provide revenues long-term. For these segments, they need to target and bring in more doctors who "own" these patients and improve their brand in these service areas through promotions, thought leader excellence, or community activities that would raise their profile. And they should further segment other patient/ service types (the elderly) in order to realize more profitable patient/ service types within current segments.

On the costs side, the hospital could realize significant decreases in variable costs by trimming down patient days, by even a slight amount on a per patient basis. While this is where the numbers and quantitative analysis is more telling, Sugar Magnolia has a social mission to be concerned with, as well as the financial mission that we've already discussed at length. Because the hospital needs to provide the best service and healthcare to its patients, it may not be able to cut down on patient/ days per patient (the costs side) while maintaining its social goal/ mission. And for this reason, several quality of care analyses should be done before initiating some of the cost-cutting initiatives that have been discussed here.



Case 6: Private Label Sales

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Brainstorming- completely qualitative

CASE QUESTION

We're going to look at a growing trend in the Consumer Packaged Goods industry. Our client is a bottled water company (e.g. Poland Springs). A major retailer (e.g. Walmart) wants our client to create a private label version of its product. In other words, in addition to our client's bottled water which they already carry, they want the client to make an additional, lower-priced bottled water which will be their own brand label, (e.g. Walmart Bottled Water).

What are the pros and cons of doing this?

INTERVIEWER BRIEFING

As it was given in the interview, this is more of a situation analysis and brainstorming exercise rather than a case to drill down on and crack. As such, there may be many more pros/cons beyond what is listed in the example below. In general, it is important to be structured and try to put the pros and cons into clear buckets. Considering the 3C's (e.g. impact on customer relationship, company's operational issues, and competitive dynamics) could also be helpful in this case.

EXAMPLE SOLUTION

Pros:

- Improved relationship with a powerful merchant (may get better shelf space and better terms on other products they purchase from us as well as cross-marketing arrangements)
- Larger production may achieve better economies of scale in both fixed costs (cost/unit to produce, esp. if certain operational procedures are synergistic) and variable costs (delivery, distribution, etc.) in addition to the potential to negotiate better terms with suppliers due to larger orders
- Huge potential revenue growth for our company (lower price but potentially high volume)

Cons:

- Cannibalization of our own private label's water sales (cost/benefit analysis of whether the volume of Wal-Mart's brand will make up for it)
- Possibility of lessening our power with the merchant as a supplier if private label takes off and supplants our own product



- Higher costs (Fixed- such as additional plant requirements due to potential capacity constraints – will the investment be worth it?)
- More complicated distribution adding additional SKU into the mix
- Potentially a different market for our product which does not work synergistically with our marketing focus (for example if our product is about prestige and image, we would be catering to a different customer segment with a lower priced product)



Case 7: Piano Market Sizing

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, Final

Content: Market sizing

CASE QUESTION

We're going to look at the sales of Pianos in the United States. What do you think annual sales (total revenues) are in the US for Pianos?

INTERVIEWER BRIEFING

Answer should show a logical way of sizing the market. You need to make some assumptions, and show your way of getting to an answer. Actual numbers aren't as important as the logic behind the method.

EXAMPLE DIALOGUE

I am thinking there are two types of piano buyers, personal households and institutions. Of those, a certain percentage already own a piano. Piano buyers, in a given year, would be divided into:

- 1) First-time piano owners
- 2) Upgrading an old piano or replacing a damaged piano
- 3) Adding a second piano (small for households, larger for institutions)

Then, I would need to estimate who these buyers might be:

Households: For example, approx. 300 Million people in the US. Avg. household is 3 (which might be high due to many single people.)

So, 100 Million households.

From there deduce how many are in the \$75,000/year income bracket, as they are most likely to purchase a piano (for example, 25% which equals 25 Million households.)

Out of those 25 Million, 5% currently own a piano already (1.25 Million Pianos).

If I assume every year, 2% of the 25 Million will buy a piano, then there are 500K pianos sold to households every year.

Follow a similar logic for institutions – e.g. colleges, universities, symphonies, Carnegie Hall, bars/businesses, etc. Assume that there are 1 million of these institutions, of which 20% (200K) would want a piano. Of that, perhaps 5% buy a piano every year, or 10K pianos sold to institutions.

Then, I would need to figure the average price paid per piano.

It is important to distinguish between new and used piano sales, as the price points are different.



In the New Piano market, I would think the low price for a piano is \$5,000 and the high price is \$15,000, so the avg price of a new piano is \$10,000.

For a used piano, the low is probably \$1,000 and new is \$5,000, so the average price is \$3,000.

I would venture that 20% of pianos sold are used and 80% are new for both households and institutions. If this is the case, 102K total new pianos are sold and 408K used pianos are sold. To make the numbers simple, use 100K and 400K. This leads you to \$1B in new piano sales and \$1.2B in used piano sales, or \$2.2B in total sales.



Case 8: Gas Stations and Convenience Stores

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Brainstorming, largely qualitative

CASE TOPIC

Our client is major global oil company that owns the whole value chain: oil rigs, refining, distribution, and retail. Our direct contact is the CEO of the global retail operation. His operation consists of 1) gas sold at the pumps and 2) the convenience stores at the gas station. Profitability of the retail operation has declined, and the CEO would like us to help figure out why and to come up with a plan for the next five years.

INTERVIEWER BRIEFING

This case is representative of many of the prepared, McKinsey round 1 cases, in which the interviewer actively walks the interviewee through a set of qualitative and quantitative questions. The interviewer should “stick to the script” of questions. To the effect that the interviewee struggles, the interviewer can assist the interviewee to get back on track.

The interviewee should be structured in answering qualitative questions and crunch through any numbers thrown his or her way, always keeping in mind how they tie back to the larger issues.

CASE QUESTIONS

Over the past fifteen years, the number of gas stations worldwide has declined by six percent.

Q: What might be the causes of this?

A: Possible answers include

- Consolidation
- Increase in dollar volume per station
- Changing population patterns (fewer rural stations)
- More stations open 24 hours (so fewer stations needed)

Set of questions re: comparative gas station profitability

It turns out there have been two other changes in the market. One, the number of gas stations with convenience stores attached has increased. Two, a major new entrant has begun taking market share. Supermarkets have begun opening gas stations in their parking lots. This is not yet a major competitor in the US, where they only have ten percent of the market, but supermarkets have 30% of the gas market in the UK, and 60% in France.



The next task is to understand whether the supermarkets have a better business model than the gas stations in this market, and if so, why. We'll use the metric of Return on Invested Capital – operating profit / invested capital.

The numbers for the UK supermarket are as follows: they sell ten million litres of gas per year at 72 cents/litre. Their cost is 20 cents/litre. They pay 45 cents/liter in tax. The convenience store's operating profit is 500,000 pounds per year. Overhead in the industry is typically ten percent of fuel sales, and that's accurate here. The capital cost is two million pounds.

We'll compare it to one of our typical gas station locations: downtown, one of our busiest locations. This location sells six million litres per year at 75 cents/litre; its cost and tax per litre are the same as the supermarket. Convenience store profit is 20% lower than the supermarket's. Overhead is still ten percent of fuel sales, and capital costs are four million pounds.

Q: What is the ROIC of the supermarket? A: (24%) (It also may come out here that the convenience store is responsible for all the profits.)

Q: Without running the numbers, what do you think our client's gas station's ROIC will be? Why? A: (Much lower – because of the cost of capital).

Q: What do you think causes that high cost of capital? A: (downtown location – supermarkets typically in the suburbs.)

Q: So, given that, what other things that drive ROIC might we be able to affect? A: (Likely can't change cost or tax – could lower price to sell more gas – could move out of the city – could attempt to increase convenience store profitability.)

Client recommendation:

The client agrees with our recommendation to focus on the convenience store, and decides to set a pilot program in 1,000 stores.

Q: What type of new products should he introduce? How would you think about what products to introduce? [Interviewee was required to come up with eight answers – near the end he was helped along with “Think about what we'd ask if he came to us with product A and product B – what would we ask to be able to decide between those two.”]

A: Some factors we could use to decide were:

- What does the existing customer want
- What products have high margins
- What can we (and the supplier) support logistically
- What can we get from existing suppliers
- What can we link to products that already sell well
- What products are needed frequently / will drive visits
- What products are durable
- What products require little shelf space (space at a premium in these stores).



In the end the decision was made to introduce hot and cold food – high-margin, low shelf-space, high-frequency (but low durability). McK is currently implementing in Europe. End of case.

Case 9: Megabank Underpenetration

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE TOPIC

Megabank is a bank that issues credit cards. New cards are sold in three main ways:

1. cross-sell to existing banking customers
2. sell to new customers via direct mail campaigns
3. distribute via private label partnerships with retailers and airline

Megabank is looking for new card member growth areas in the United States. Its Hispanic market penetration is low compared to comparable banks' penetration rates. That group is a fast growing ethnicity and the bank wants to capitalize on it.

What is the current problem (i.e., what are the possible reasons for the underpenetration in the Hispanic market?)? How should the bank move forward?

INTERVIEWER BRIEFING

Recommended approach:

This case is representative of many of the prepared, McKinsey round 1 cases, in which the interviewer actively walks the interviewee through a set of qualitative and quantitative questions. The interviewer should "stick to the script" of questions. To the effect that the interviewee struggles, the interviewer can assist the interviewee to get back on track.

The interviewee should be structured in answering qualitative questions and crunch through any numbers thrown his or her way, always keeping in mind how they tie back to the larger issues.

CASE QUESTIONS

Q: What might be wrong? Hypothesize.

A: Discuss briefly the potential of each of the following to affect the penetration rate:

- Product Definition – Does the card, as it has been defined meet the needs of the customer?
- Pricing of Credit Card Terms – Are the fees and rates on par with other comparable cards?
- Marketing/Advertising – Are the messages properly directed to the audience (both content and distribution)?
- Channel Partners – Does our target audience shop at / eat at / buy from our partners?



- Internal Sales Messages/Incentive Structure – Are the sales messages correctly structure to entice our potential customer? Is our internal sales force (i.e., teller and desk personnel) trained and incented properly to promote the card? Discuss in detail (i.e., what might be misaligned, how that might affect adoption rates, etc.).

Q: Calculate the number of additional members Megabank wants to add based on the following information:

- There are 40 million Hispanic people in the US
- 3/8ths of them are too young to have credit cards
- The average customer is worth \$180 to the bank over the course of his life
- Due to decreased acquisition costs, the average Hispanic customer is worth 10% more
- Currently, the bank's penetration rate is 10% (of valid customer prospects)
- They want to get to a 30% level over 5 years

A: 3/8ths of 40M are too young, so 5/8ths of 40M are valid customer prospects. That translates to a market of 25M people.

They currently have 10% of 25M = 2.5M

They want 30% of 25M = 7.5M

They need 5M additional members over 5 years

Q: How much is that additional market share worth to Megabank (or how much would Megabank be willing to spend on that additional market share)?

A: If the average customer lifetime value is \$180, but the average Hispanic customer is worth 10% more, each customer is worth \$198. Rounding that to \$200, the total value of 5M extra member is \$1B.

Q: Cross-selling to branch customers is significantly below the industry norms (Average = 15,000 to 20,000 per month; Megabank = 5,000 per month). What might be the reason?

A:

- Product – Is the Megabank product different from competitors' products? Interviewer: *NO*
- Pricing – Are the fees and rates different than other comparable cards? Interviewer: *NO*
- Customer – Are we targeting a fundamentally different audience? Interviewer: *NO*
- Channel Partners – Are the distribution channels misaligned? Interviewer: *NO*
- Marketing/Advertising – Is there something wrong with our sales mechanism? Interviewer: *Let's investigate*



Q: Megabank uses direct mail campaigns to solicit new card members.

- The average response rate for the non-Hispanic population is 1%.
- Megabank's historical response rate from Hispanic prospects is 3%.
- The bank is planning to target 15M potential customers with each of 3 mailings this year.
- It expects that after the first mailing, the response rate will drop by $1/3^{\text{rd}}$ in each of the subsequent mailings.
- The bank has a conversion rate of 45% of respondents.

How many new customers should the bank expect after the third mailing?

A: $3\% * 15\text{M} = 450,000$ from first mailer

$450,000 - 1/3 * 450,000 = 300,000$ from second mailer

$300,000 - 1/3 * 300,000 = 200,000$ from third mailer

$950,000 * 45\% \text{ conversion} = \sim 450,000$ new customers

Q: You rounded 950,000 to 1,000,000. Would you expect the actual number of new customers to be more or less than 425,000 and how do you calculate that (in your head)?

A: Slightly more since 10% of 950,000 is 95,000; therefore 40% is $4 * 95,000$ or 380,000. Add 5% of 950,000 (or half of 95,000) which is 47,500 to 380,000 to get 427,500. That would be somewhere between 2M and 2.5M over five years.

Q: You bump into the SVP of Sales (related to the credit card business) in the hall and he asks "How does it look?" How do you respond (1 minute only)?

A: We are still crunching the numbers, but based on your current market position, your goal of increasing penetration by 20%, and historical conversion rates for direct mail campaigns, our initial estimates suggest that you will fall short of your goal by mainly relying on that method of acquisition. In fact, it will only get you about half way to your goal. We need to discuss other measures to increase penetration of the Hispanic market. Specifically, we need to look at your sales force compensation structure, training and specific sales and marketing messages. Let's plan to review our formal recommendations later in the week.

Case 10: Heartcorp

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE TOPIC

Heartcorp is a medical devices company. They produce cardiovascular stents. They have developed a revolutionary product that is positioned to replace the current products in the market. The product, called Device X, is the first of its kind

Heartcorp wants to launch Device X in Europe in the near future and then bring it to the U.S. in 6 months. They are 1 year ahead of its competition (with regard to R&D of the product).

INTERVIEWER BRIEFING

Recommended approach:

This case is representative of many of the prepared, McKinsey round 1 cases, in which the interviewer actively walks the interviewee through a set of qualitative and quantitative questions. The interviewer should “stick to the script” of questions. To the effect that the interviewee struggles, the interviewer can assist the interviewee to get back on track.

The interviewee should be structured in answering qualitative questions and crunch through any numbers thrown his or her way, always keeping in mind how they tie back to the larger issues.

CASE QUESTIONS

Q: How do you determine what price to charge for Device X? What are the issues that need to be considered?

A: Three areas should be explored to determine the price:

1. Current price and rationale for current price (value-based or cost-based)
2. Benefits of new drug vs. old drug in terms of decreased side effects or repeat procedures
3. Buyer’s willingness to pay

Additional items that could be considered:

- Cost of Device X
- Total R&D cost (if not considered sunk)
- Any customer acquisition cost
- Cost of overhead and sales force



Q: What is the “Cost Neutral Point”- the point at which the cost of the new product equals that of the old product- given the following data:

- Hospital Cost of an operation = \$2500 per patient
- Old product cost = \$500/unit
- Number of units needed per operation = 2
- Frequency of repeat procedure (using existing product) = 50%
- Frequency of severe complication resulting in open heart surgery = 30%

Is there any more information that you would need to calculate this?

A: Need more information:

- What is the cost of open heart surgery operation? *\$15,000*
- What is the frequency of repeat procedure using Device X? *0%*
- What is the frequency of severe complication using Device X? *5%*
- How many units of Device X will be necessary per operation? *2*

To determine how much is the new device is worth:

<u>Item</u>	<u>Old</u>	<u>New</u>
Hospital cost	\$2500	\$2500
Product cost per operation	$\$500 \times 2 = \1000	$2 \times X$
<i>Subtotal</i>	$\$3500$	$\$2500 + 2X$
Repeat procedures	$50\% \times \$3500 = \1750	$\$0$
Severe complication	$30\% \times \$15K = \4500	$5\% \times \$15K = \750
<i>Total</i>	$\$9750$	$\$3250 + 2X$

Value of Device X = $(\$9750 - \$3250) / 2 = \$3250$ (cost neutral point)

Q: What might factors might allow Heartcorp to price the new product above cost neutral point)? What needs to be considered?

A:

- Risk / Malpractice Insurance costs,
- Value of reduction in pain (to patients),
- Higher success ratio (without repeat procedure and/or severe complication),
- Cost savings of keeping fewer Device X's in inventory, etc.

These and other factors might allow us to price Device X above the Cost Neutral Point.

Q: The company has decided that it wants to sell Device X at a premium above the cost neutral point, but a survey of potential customers (Hospital Purchasing Departments) showed that they are only willing to pay \$2000/unit. Now what would you recommend?



A: Is the \$2000/unit figure a single data point or an average? *Average across many customers surveyed*

One of two things:

1. Manage Heartcorp's expectations that they should really expect something close to \$2000/unit
2. Increase potential customers' "willingness to pay"

Q: How can we increase "willingness to pay"?

A: Two thoughts:

- Communicate the benefits (both "soft" (e.g., decreased pain or frequency of re-operation) and "hard" (financial)) to the additional stakeholders (i.e., patient advocate groups and insurance payers) in the decision. Work with them to "pressure" the potential buyers of Device X to spend the additional money to realize the added benefits of the new hardware.
- Publish articles about the efficacy of the new device in reputable journals (e.g., JAMA, New England Journal of Medicine, etc.) and use those to convince doctors to pressure hospital administration to increase "willingness to pay" for Device X.

Case 11: Credit/Debit Card Processor

BACKGROUND

Firm: McKinsey & Company

Round: 2005 Summer, First

Content: Qualitative and quantitative

CASE TOPIC

Pierce Processing is a credit/debit card processor. It provides outsourced services to credit/debit card issuers. The issuer pays a percentage of transaction fees to Pierce for providing the following services:

- credit authorization,
- fraud detection,
- accounting and reporting.

Issuers outsource this function to companies like Pierce because of the high cost of maintaining their own IT systems to perform this function. Pierce has hired us to find ways to sustain its double digit growth that the company has enjoyed for the past 20 years.

INTERVIEWER BRIEFING

This case is representative of many of the prepared, McKinsey round 1 cases, in which the interviewer actively walks the interviewee through a set of qualitative and quantitative questions. The interviewer should “stick to the script” of questions. To the effect that the interviewee struggles, the interviewer can assist the interviewee to get back on track.

The interviewee should be structured in answering qualitative questions and crunch through any numbers thrown his or her way, always keeping in mind how they tie back to the larger issues.

CASE QUESTIONS

Interviewer: *What steps could Pierce take to keep a high growth rate?*

Interviewee: What is Pierce’s current market position in the credit card market?

Interviewer: *Pierce currently serves the 10 of the top 20 issuers of the country. The 20 top issuers comprise 90% of the credit card market. The 10 of the top 20 that are not Pierce’s customers either use Pierce’s competition or perform the function in-house.*

Interviewee: How fast is the credit card market growing?

Interviewer: *The number of credit card transactions is growing at 3% annually.*

Interviewee: How does Pierce price currently in the credit card market?



Interviewer: *Pierce earns a percentage of the transaction fee of each transaction.*

Interviewee: Is there any other revenue stream?

Interviewer: *Someone has suggested that Pierce could license its software as another business model, in addition to being the outsourced processor. In this case, the issuer would license Pierce software and process the transactions in-house. Pierce would get an upfront license fee and ongoing maintenance fee. What should Pierce consider in evaluating the software license model?*

Interviewee: Possible answers:

- Danger of cannibalization,
- Impact of new revenue source- upfront cash and recurring revenue stream
- Added cost of maintenance and support of this business model.

Interviewer: *Assume that if Pierce offered the software license model, 50% of its existing customers will switch to this model. 50% of the remaining 10 issues that are not currently Pierce customer will sign with Pierce. The revenue of providing outsourced processing is \$1M/year. The upfront fee of license model is \$3M and \$300K in annual maintenance fee. Calculate the impact on revenue for the next 5 years if Pierce offers this new licensing model (ignoring time value of money).*

Interviewee: Revenue is \$70M under new plan for the next 5 years, compared with \$50M if the software license model is not adopted.

Interviewer: *Good. What else would you like to know in order to reach a recommendation?*

Interviewee: What is Pierce's current market position in the debit card market?

Interviewer: *Pierce has 5% market share.*

Interviewee: Who are Pierce's main competitors in the debit card market? How much market share do they have?

Interviewer: *The market is very fragmented. No player has more than 10% market share. What can Pierce do to gain more market share in the debit card segment?*

Interviewee: Possible answers: acquire competitors, set up a partnership with a competitor, use innovative marketing schemes (e.g. incentives)

Interviewer: *There is another segment of the market, store value segment, where Pierce currently has no offerings. This is the market of store-specific gift cards, i.e. Starbucks card or GAP gift cards. This market is projected to grow at 50% annually over the next 10 years. What are some of the issues to consider in determining whether Pierce should enter this market?*



Interviewee: Cost of offering this new service vs. potential revenue, difficulty in setting up partnerships and Pierce's experience with this, complexity of a revenue-sharing scheme, "stickiness" of store-aligned cards vs. regular cards, potential cannibalization of regular cards

Interviewer: *Suppose we know that the total market of store value is \$1B today. The transaction fee is 1% of the total value. Processing firms, like Pierce, would earn about 10% of the transaction fees. What is expected revenue from this new segment this year?*

(Note to interviewer: See if the interviewee asks for a market share number. If asked, Pierce expects 30% market share.)

Interviewee: So Pierce would expect revenue of \$300K ($\$1B * 1\% = \$10M$ total transaction fees, X 30% market share = \$3M Pierce-related transaction fees, X 10% = \$300K Pierce take of the transaction fees).

Interviewer: *Suppose it costs \$1M to build the system required to process store cards. Assume 5-year useful life, what's Pierce's expected profit (accounting profit) in the first year?*

Interviewee: Answer: $\$300K - (\$1M/5) = \$100K$



Case 12: European Motorcycles

BACKGROUND

Firm: Bain

Round: 2006 FT, Final

Content: Qualitative and quantitative

CASE TOPIC

Our client is a private equity firm that has just purchased a motorcycle manufacturer based in Germany that makes sporty bikes. Currently they supply motorcycles to Europe but have no presence in the US. They are looking to expand the business into the US but are unsure of how to do this. They are considering two models: one is selling through existing dealerships in the US, the other is building a flagship dealership and selling that way.

Our client wants to consider a few issues. First, is this expansion a good idea? Second, which model would be the best to pursue?

As a starting point, tell me a little about the high level pros and cons of building a flagship dealership versus selling through existing channels.

INTERVIEWER BRIEFING

Recommended Approach:

High level, you're going to need to hit the numbers right when they're presented and be able to step back and synthesize into some sort of meaningful recommendation. Interviewer should keep asking the interviewee to summarize and back-up assumptions.

Interviewee should keep in mind that this is a PE shop and wants to flip the company quickly – this is a common theme through Bain cases.

Key Facts:

Flagship:

Retail Price: \$20K

Cost: \$10K

Cost to set up flagship: \$3 million

Existing Channel:

\$250K start up cost

Retail Price: \$20K

Distributors require 33% gross margin, then there will be a 3% allowance of the sales price that will be reserved for return, selling costs, etc.



EXAMPLE DIALOGUE

Interviewee: I would think that opening a flagship dealership would have a lot of pros. First, it would probably allow for higher margins as you would own the distribution. Second, you could train the sales people to sell the right way, position the product correctly, etc. You would also then control your customer data, ability to control marketing messages in the dealership, etc. On the downside, there is the capital investment, the fact that existing dealerships have relationships with customers already in place, distribution channels already in place, and local market knowledge. Your brand name is also not well-known in the States, so opening a flagship dealership might be difficult.

Interviewer: *OK, that makes sense. What would you think about the profitability?*

Interviewee: I would assume that owning your own dealership would be more profitable in the long run, but short term it might be better to go with the existing channels b/c of the capital investment. Given that this is a PE firm, they probably don't want to invest a ton of capital or hold for too long.

Interviewer: *That sounds good – let me give you some numbers and let's see if they match your hypothesis:*

Flagship:

Retail Price: \$20K

Cost: \$10K

Cost to set up flagship: \$3 million

Existing Channel:

\$250K start up cost

Retail Price: \$20K

Distributors require 33% gross margin, then there will be a 3% allowance of the sales price that will be reserved for return, selling costs, etc.

Interviewer: *What's the breakeven in both cases?*

Interviewee: The breakeven in terms of a flagship store is 300 bikes ($\$3,000K / [\$20K - 10K]$). For the distributor model, you assume a 33% gross margin, so 33% of 20K is 6.6K. That means we still get 13.3K. With a 3% allowance, that means we roughly take in 13K for every bike. That's 3K profit. $\$250 / \3 is about 83 bikes.

Interviewer: *Right. What does that tell you?*

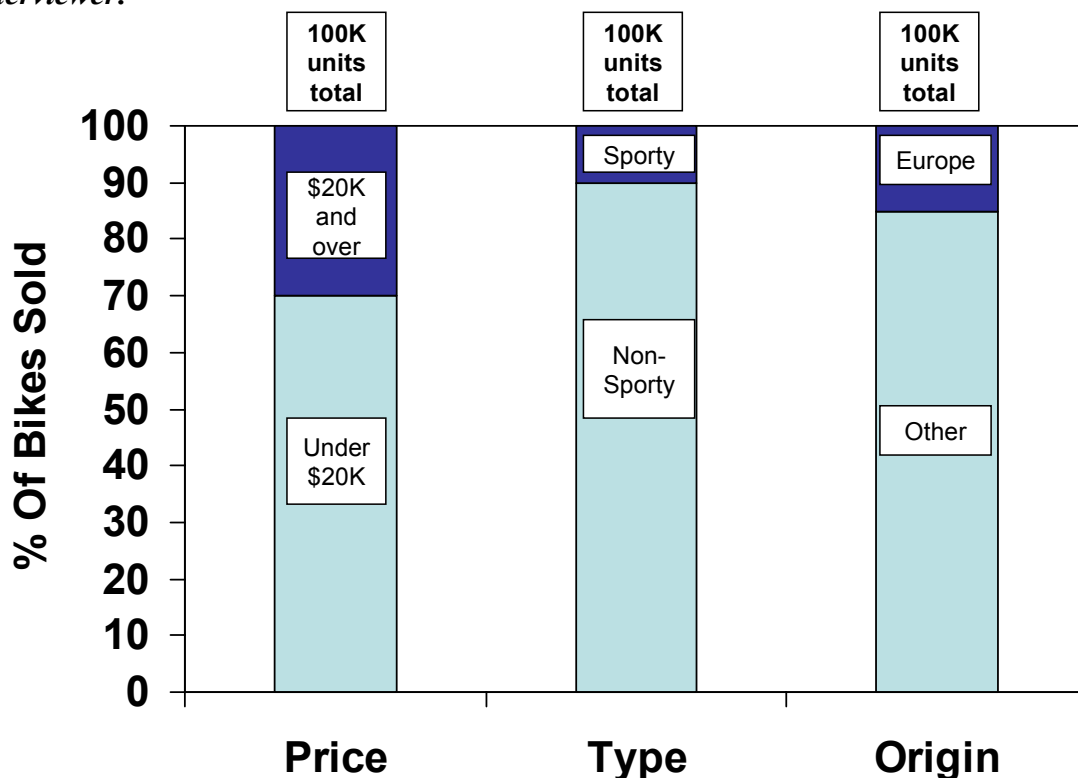
Interviewee: That our breakeven is lower in the existing distributor model. It also tells me that in terms of investing in the long term the flagship would be MUCH more profitable.



Interviewer: So how conceivable are these numbers?

Interviewee: I would think selling 83 bikes should be fairly easy in a year. 300... it also sounds reasonable.

Interviewer:



(Presents chart – chart shows all sorts of things but relevant info is that 100K bikes are sold in the US every year, and it's broken down 3 ways. First by country of origin, next by price, third by type. Turns out 15% of bikes are from Europe, 30% are high end (20K or higher), and roughly 10% are sporty.)

Now what do you think?

Interviewee: (takes time to think) Alright, so now I know 100K bikes are sold. So 15K of them come from Europe, 30% of those or 4.5K are high end, and 10% of those are sporty. That's 450 bikes. So maybe 300 would be a stretch, and 83 might even be tough if there are strong competitors.

Interviewer: I would argue that the country of origin doesn't matter. A sporty 20K bike is a sporty 20K bike, right?

Interviewee: I'd actually disagree. Think about automobiles – some people buy German cars b/c they want to buy German cars, nothing else. Or think about beers. Some people only like



Belgian beers for no reason except that their Belgian – it's an image thing, right? (Note: This was not well received by the interviewer.)

Interviewer: *OK, there's no right or wrong answer. Assume that I'm right and the market is much bigger. Now what?*

Interviewee: OK, then it seems feasible. So short term the better play is to use distribution centers that are already in place, but long term it would be to build a flagship.

Interviewer: *And does the fact that it's a PE client have any bearing here?*

Interviewee: Well, they'll want to turn it over quickly and not pour cash into it, so they'd likely go with existing channels and not build. They also might want to examine why they're thinking about the US – maybe they can grow in other parts of Europe more efficiently and profitably.

Interviewer: *Great.*

Case 13: Street Sweepers

BACKGROUND

Firm: BCG

Round: 2006 FT, Final

Content: Mainly qualitative

CASE QUESTION

Our client is a large industrial conglomerate that has \$60-70 million in annual revenues and is extremely profitable. They have engaged BCG to examine one of their business units that is underperforming. The unit manufactures and distributes street sweepers (zamboni/lawnmower-like machines that a person would ride and which uses two large rotating brushes to sweep up the street as it moves by) and has been doing so for 20 years. While never a standout division, until recently it had always been profitable. 4-5 years ago the profit margins started to fall and it is currently just above breakeven.

Management of the conglomerate had made a decision 2 years ago that they did not want to invest in new features for their product lines and has the approach that they don't want to invest significant amounts of capital now (unless they can be convinced otherwise).

BCG's task is to analyze the unit's performance and recommend to the management of the conglomerate what should be done. Tell me some of the areas you'd consider looking into here knowing that we only have a few weeks to finish this engagement.

INTERVIEWER BRIEFING

The interviewer did not allow time for development of a framework – that isn't the point of the case. Instead, the interviewer simply starts asking questions and they should be answered as thoughtfully as possible on short notice. Being thoughtful but remaining structured is vital here: sticking to profits = revenue – costs and drilling down on costs and revenues. This is a great case b/c if you understand basic drivers- it's completely rational.

EXAMPLE DIALOGUE

Interviewee: That's a tight timeline, so let's focus on big areas. So profit is revenues minus costs. Can you tell me about recent trends with regard to costs?

Interviewer: *Sure. A quick look at the numbers showed costs have been fairly constant over the past 20 years, only increasing with inflation.*

Interviewee: So I'm thinking I'll then move into revenue. How has that stood up?

Interviewer: *Revenue has fallen steadily over the past 5 years. Why might this be happening?*



Interviewee: It might be happening because of trends in the market, changes in customer preferences...

Interviewer: *Since you mention it, who are the customers?*

Interviewee: I suppose they would be municipalities, governments, etc. I don't see this as being something for individual use.

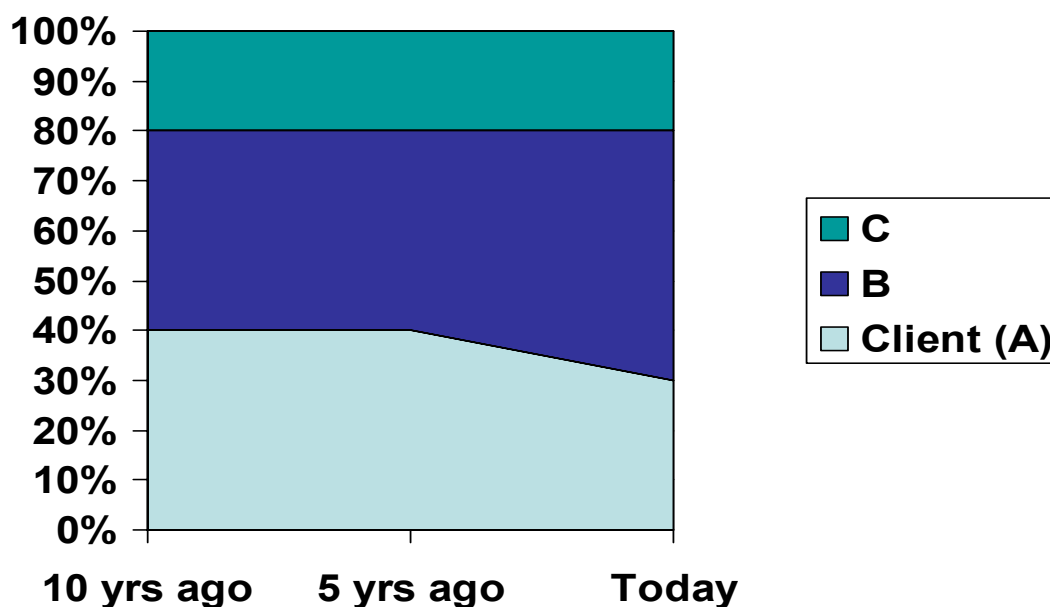
Interviewer: *Usually not. That's right, it's mostly state and local governments. So we looked into the customers and they are keeping with past trends, replacing their street sweepers every 4-7 years, and the number of municipalities purchasing street sweepers has remained constant. What else might be driving revenues down?*

Interviewee: Prices could be coming down...

Interviewer: *Prices have actually remained constant.*

Interviewee: So how about competition? Have new competitors entered the market or stolen market share?

Interviewer: *There have been no new competitors. These are what sales have looked like 20 years ago to now. What do you want to know after looking at this?*



Interviewee: It looks like B has taken share from us. I'd like to know what B is doing differently. And for that matter what C is doing differently so as to be unaffected.



Interviewer: *About 5 years ago, B introduced a new technology that used air vacuums to clean streets instead of the mechanical collecting methods that had always been used in the past. These air machines are more effective at picking up small debris like sand and small litter, and work more quickly and efficiently than the mechanical ones previously offered. Price points are about the same. B still sells mechanical machines, but the drop in our market share was directly related to the new air offering.*

Interviewee: So how about C?

Interviewer: *C makes machines that are far more heavy-duty, it's really a different type of offering.*

Interviewee: So we don't compete with C in reality?

Interviewer: *No, but we could. Do you think we'd want to?*

Interviewee: Not likely without knowing much about the market. They seem to have been stable and have an expertise, so unless we can offer something new to their customers I'd guess they'll defend their position in a niche market at all costs.

Interviewer: *Good, that's the conclusion we quickly came to on C. Now, what other information do you need to recommend something for our client?*

Interviewee: I need to know if we can replicate the air technology.

Interviewer: *We can, but it will take 2 years and cost \$25 million.*

Interviewee: Since you said management does not want to make a significant investment, that seems unlikely. So other than that, what trends do we expect from this market going forward?

Interviewer: *What would you think?*

Interviewee: (Pause) I guess the encouraging news is that we're still selling despite the new technology and the similar price point. So do municipalities need both?

Interviewer: *Good observation. In fact they do need both. So we broke the country into 7 regions, all of which needed the same amount of street sweepers overall. In the north, on average there was a 22/8 ratio of mechanical to air sweepers needed. In the south, it was 22/18 (these numbers do not include C's models). For the country, what % of street sweeper sales were going to be that are mechanical?*

Interviewee: Roughly 66% (It's actually 64%).

Interviewer: *Yes. So why would municipalities need mechanical sweepers if air ones are more efficient, and what does this tell you?*



Interviewee: I'd think they'd need mechanical ones to deal with larger, more solid things on their streets. It looks like in the south they can use more air machines which would make sense b/c they'd have sand and small debris. In the north you might get larger rocks, chunks of ice, etc. This tells me the market isn't going to zero, so maybe the company can simply protect the market share that it has, scale back production to the point where the market will be in equilibrium between air and mechanical sweepers, and keep decent margins going forward.

Interviewer: *Right, so that's exactly what we recommended. Once we recognized these trends, we also looked for a strategic buyer, and in fact the conglomerate sold the unit to a foreign company that already had air technology developed. That was outside the scope of this case, though – just an interesting follow-up.*



Case 14: Eye Surgery

OVERVIEW

Firm: BCG

Round: 2006 FT, First

Content: Market sizing, qualitative and quantitative

CASE QUESTION

Our client is a manufacturer of equipment for eye surgery. Specifically, the machines measure deficiency in eyes, and the company also produces lasers for post-operational procedures and adjustments. They don't actually make the lasers or devices used for Lasik – rather, they are complementary products for this procedure.

The global market for these devices is growing, but at a declining rate. As a result, the client wants to get into a higher growth area, so they are looking at acquiring a company that makes inter-ocular devices. These devices are used instead of Lasik but with similar effectiveness, and they are used for two major categories of patients:

- 1 – patients with cataracts
- 2 – refractive surgery (to correct near or far sightedness)

How would you approach this opportunity? What would you look at?

Additional information provided during questioning

Interviewee asked about the specifics of what the machines were so as to consider synergies between the two companies and product offerings. There would be significant synergies and that is a component of answering the case.

INTERVIEWER BRIEFING

Recommended Approach

Given that we're looking at a company with an existing product line that is exploring moving into a related product line, we need to understand any links between the two. It is vital that the interviewee demonstrates his acknowledgement of the risks of cannibalization and the benefits of synergy between the old and new lines. Also key is to show an understanding of some of the basics of M&A. High level, a framework looking into internal factors of both the target and acquirer (such as culture, finances, and the synergies there might be between the two), external factors such as market trends and competition, and customer factors (both doctors and patient segments) is necessary. The interviewee should also remember the significance of the valuation of the target- is it worth the asking price.

This case is fairly simple if you hit the numbers – take your time and get them right. The overall framework was very helpful as interviewee was able to reference it multiple times during a fairly



focused case discussion. The key is identifying that there will be different types of customers for each offering, so suggesting IDing customer segments up front seemed to be a major plus.

Key Facts

- US population is roughly 300M
 - 75% of the US population over 65 has cataracts
 - US population is evenly distributed over 80 years, the same number of people are each age
 - When someone turns 65, they have a 75% chance of getting cataracts, and if they don't get it immediately they will never get it
 - 1/3 of the population is near-sighted and 1/4 of the population is far-sighted-> 175M people need vision correction of some kind
- There are government caps on pricing for cataracts surgery and that there is substantial competition from major national players.
- The refractive market is still very fragmented and growing rapidly – 1.5M surgeries/year will grow to 3-4M as procedures become safer. Also, the patient pays 10x as much for refractive surgery as a cataracts patient would pay.

EXAMPLE DIALOGUE

Interviewee: I would look into internal factors of both the target and acquirer (such as culture, finances, and the synergies there might be between the two), external factors such as market trends and competition, and customer factors (both doctors and patient segments). Related to all of these would be the valuation placed on the company. If we could, I'd like to start with drilling down on the customers.

Interviewer: *OK, I like that. So let's talk about the cataracts patients. If I were to tell you that 75% of the US population over 65 has cataracts, how many potential patients are we talking about?*

Interviewee: Well I know that 12% of the population is 65+, so let's call that 10% for simplicity. 10% of 300 million is 30 million. 75% of that is 22.5 million. But some of those people might already have had surgery.

Interviewer: *Good point. And it gets a little dicey because the segment would be skewed towards 65. So here is a simplifying assumption – assume the US population is evenly distributed over 80 years, the same number of people are each age. When someone turns 65, they have a 75% chance of getting cataracts, and if they don't get it immediately they will never get it. What's the market size thinking this way?*

Interviewee: OK, so we have 300 million people over 80 years. That's 3.75 million people in each year age bucket. So it would be 3.75 million people turning 65 every year. If 75% of them get cataracts, that's... roughly 2.9 people a year. Plus some percentage of the population already over 65, I'm thinking right around 3 million people a year.



Interviewer: *Does that make sense?*

Interviewee: I don't know a lot about cataracts, but it seems to. I'm not sure all of those people currently get laser eye surgery currently, though.

Interviewer: *Right. OK, now let's turn our attention to the refractive surgery market. So your research tells you that 1/3 of the population is near-sighted and 1/4 of the population is far-sighted. Assume that those numbers already include those who've had their vision corrected. How many people are we talking about for the potential market size?*

Interviewer: (Works out $4/12 + 3/12 = 7/12$; $7/12 * 300$ million people total = 175 [shortcut: $1/4$ of 100 = 25, $* 7 = 175$]) 175 million people.

Interviewer: *Right. And it turns out that it translates to 1.5 million people a year actually getting refractive surgery. So if we acquire this company and can position it as a cataracts provider or a refractive surgery provider, which should we position it as? (note: the machinery would be slightly different, enough so that it would be beneficial to go after one market or the other).*

Interviewee: OK, so I know that the cataracts market is around 3 million a year and the refractive market is 1.5 million a year. But I don't know anything about profitability so I can't really say. Can you tell me a bit about the markets?

Interviewer: *What do you want to know?*

(Key information: There are government caps on pricing for cataracts surgery and that there is substantial competition from major national players. On the other hand, the refractive market is still very fragmented and growing rapidly – 1.5 will grow to 3-4 as procedures become safer. Also, the patient pays 10x as much for refractive surgery as a cataracts patient would pay).

Interviewee: So based on what we just discussed I'd like to target the refractive market.

Interviewer: *Is there anything else you would want to know before making a decision to buy the company?*

Interviewee: I'd need to know more about the financials to give a clear answer. I'd also need to better understand the synergies and how they'd be perceived in the market. However, it looks promising given our examination of the market segments.

Interviewer: *Excellent.*



Case 15: Meditest

OVERVIEW

Firm: McKinsey and Company

Round: 2006 FT, First

Content: Market sizing, qualitative and quantitative

CASE QUESTION

Our client is called Meditest – they make equipment for blood sample test labs. Specifically, their machines are used to test blood samples for their glucose levels, meaning they are used to monitor diabetes. Their machines are very expensive, often costing more than \$20K, and they must be operated by professionals. They also make the consumables that go in these machines. Meditest has a presence only in Europe.

The market for these large machines is growing very slowly and Meditest is looking for more growth. They have designed patient-operated machines that can do simple glucose level tests. These devices are small and portable, are easy to operate, and do not need much in the way of extra consumables.

Meditest is considering launching this device, called glucore, in the UK. They have retained McKinsey to examine what considerations they should have in this launch.

INTERVIEWER BRIEFING

Recommended Approach

The first step is to establish a framework. Ideally, the interviewee would take into account internal and external factors, the characteristics of the customers, and why UK?

- Internal considerations: capabilities, capital, culture
- External considerations: competition, regulation, market trends
- Customers: market size, segmentation, preferences, payment assistance
- Why UK?: why not other places, why not other value-added services for their core product?

Once you get through the framework in a McKinsey case, it's more or less listen and respond so try to think about the implications of what you're being told and analyzing.

This case represents a mix of numbers and detail analysis, then the need to step back and consider what the numbers are telling you. Take a moment to think about the numbers in context and it should be fine.

Key Facts

- Size of the market is 108K people/year
 - Population of the UK is 60 million.



- 20% of that population is over 65 years old.
- Of those over 65 years old, 5% are currently diabetes sufferers.
- Of those under 65, 1% of the population are currently diabetes sufferers.
- 10% of the population buys glucoire every year
- Revenues will be 2.16M pounds/year in 4 years
 - Price point on these machines today is 120 pounds.
 - The marketing department project that the price will fall by 50% in 4 years (60 pounds)
 - Total market will grow by 33% in terms of people with diabetes (144K)
 - In four years we will have acquired a 25% market share (36K)

EXAMPLE DIALOGUE

Interviewee: I would look at internal and external factors, the characteristics of the customers, and why the UK vs. elsewhere

- Internal considerations: capabilities, capital, culture
- External considerations: competition, regulation, market trends
- Customers: market size, segmentation, preferences, payment assistance
- Why UK?: why not other places, why not other value-added services for their core product?

Is there anywhere in specific you'd like me to start?

Interviewer: *Let's start with the customers. You said you wanted to think about the market size for this product. How would you do that? What would you need to know?*

Interviewee: Well, I'd need to know the percentage of the population with diabetes, the total population, then the percent that could afford this item. I'd need a price as well if you wanted the market size in pounds.

Interviewer: *Let's just do it in terms of people and add in the pounds later. So the population of the UK is 60 million. 20% of that population is over 65 years old. Of those over 65 years old, 5% are currently diabetes sufferers. Of those under 65, 1% of the population are currently diabetes sufferers.*

Interviewee: Through my calculation I get 1.08M people ($1/5$ of 60M = 12M; $5\% * 12M = 120K * 5 = 600K$. $1\% * 48M = 480K$. $600K + 480K = 1.08M$)

Interviewer: *Right, now what else do you need to know?*

Interviewee: I need to know how many people will buy this device.

Interviewer: *So assume that 10% of the population buys every year. How many people is that?*



Interviewee: 108,000

Interviewer: *Does that sound like a lot to you?*

Interviewee: Not really. We're talking about a company that sells machines for over 20K apiece, so without knowing anything about the price of this new product it seems like this is a pretty small total market to be going after.

Interviewer: *Alright, we'll come back to that. Assume the price point on these machines today is 120 pounds. The marketing department project that the price will fall by 50% in 4 years. Also in 4 years they assume the total market will grow by 33% in terms of people with diabetes. In addition, in four years we will have acquired a 25% market share. What will our revenues be in 4 years?*

Interviewee: The price will fall to 60 pounds (0.5×120 pounds), and the number of purchases in a given year will grow to 144,000 ($108,000 / 3 = 36,000$). Our market share is 25%, so there will be 36,000 people buying ($144,000 / 4 = 36,000$) at 60 pounds per person. That brings me to a revenue number of 2.16 million pounds.

Interviewer: *Good. Now how does that number sound to you?*

Interviewee: It still sounds a little low for the type of growth I believe this company is looking for.

Interviewer: *So what are some levers you might pull to increase that number?*

Interviewee: There are a number of drivers. First there is the population size – you might want to include other countries or markets to increase the overall field you're competing in. From market share, you could try to position yourself differently, maybe as the more convenient option or the most trusted one. Or you could try to keep prices high. I'd think the most logical would be to try to hit more people overall.

Interviewer: *You mentioned price. How would you think about pricing this product?*

Interviewee: I would think about the value add to consumers in terms of the convenience offered and money saved. If this means they don't have to go to the physician then they will be saving time and money.

Interviewer: *Do consumers always pay for their trips to the doctor?*

Interviewee: No. Often insurance would pay, so they might be willing to do some sort of a co-pay for devices like this. The devices might also prompt people to keep better tabs on their blood-sugar levels which insurance companies would like a lot. After all, the major cost to insurance companies is the catastrophe, so anything cutting down those chances would probably be a good investment.



Interviewer: *Right. So if you had to summarize what we've looked at so far...*

Interviewee: We've looked the current size of the market in the UK for consumer blood-sugar monitoring devices, as well as the growth potential in four years. There seem to be lots of benefits to the customers and to insurance companies meaning the product should be accepted if effective. However, it seems that a launch just in the UK has limited revenue potential.



Case 16: Magazine Growth

OVERVIEW

Firm: Marakon Associates

Round: 2006 FT, First

Content: Brainstorming, qualitative

CASE QUESTION

Our client is a large media conglomerate, let's call it 4-star media. 4-star functions mostly in online and print media, and within those in 3 major segments:

- 1) Enthusiast Media – something like Hot Rod Magazine, Snowboarder Weekly, or S&M for the elderly.
- 2) Consumer Guides – Motor Trend, Consumer Reports, etc.
- 3) Business News – Fortune, WSJ

We're going to be focusing mostly on the business news segment today. It's called Prime Inc. and is broken down into print and online media itself. There is a website but it basically just reproduces the news that's in the magazine. The business news segment has seen declining revenues over the past 5 years after decades of very stable growth. We've been hired to consider two things- first, we need to determine why revenues have been declining and second, we need to recommend some alternatives to stimulate growth.

INTERVIEWER BRIEFING

Recommended approach

High level, the interviewee should suggest a framework looking into internal factors such as a change in the price structures or a change in the product, external factors like competition or market dynamics, and customer factors like segmentation or preferences. The interviewee should be given the opportunity to brainstorm some ideas for growth, then follow the lead of the interviewer.

This case is straightforward if you can talk about the relationship between the end reader and the advertiser. If practicing with this case, try to get the interviewee to think outside the box as there was a lot of 'idea generation' through the course of the case. An understanding of the publishing industry is helpful but not necessary, just ask a lot of questions if it's new territory.

Key facts

As the interviewer, you should recognize that most of the revenues from the magazine business come from advertising as opposed to subscriptions. You also will know the results of some prior consumer research showing that there is low brand awareness from target customers who don't already subscribe and former / current readers don't feel the content is as insightful. The average reader is now 10 years older than they were 5 years ago. Of the top 5 magazines doing business



news, Prime Inc has the lowest share of that demographic and the highest price per ad per 1000 readers.

EXAMPLE DIALOGUE

Interviewer: *What would you look at in terms of addressing the first issue (revenue declines)?*

Interviewee: I would like to explore internal factors such as a change in the price structures or a change in the product, external factors like competition or market dynamics, and customer factors like segmentation or preferences.

Interviewer: *That makes sense. So I can tell you right away that there was no change in pricing structure or the product itself, so that takes care of internal factors. Let's talk about the customers. What would you want to know about them?*

Interviewee: I'd like to know who they are, what the target is, what they think of the product...

Interviewer: *Which customers are you talking about?*

Interviewee: I'm talking about the people who actually buy the magazine... but those aren't the only customers. There are also ad sales in any media business like this one.

Interviewer: *Right. And off the top of your head, which would you say contributes more to total revenue, subscription revenue or ad revenue?*

Interviewee: To be honest I'd have to say advertising revenue only because I know television relies on it so heavily.

Interviewer: *In fact it is advertising revenue, and it's about a 3:1 ratio. So does that mean we should focus more heavily on advertisers than the subscribers?*

Interviewee: I'd actually think the opposite. I'd think that we'd want to focus on building a loyal, desirable subscription base that advertisers want. My general feeling is that if we have the demographics that the advertisers want to hit, they will not only come to us, but it fact may be willing to pay a premium in some cases. I wouldn't see focus on advertisers as being all that useful if we can't deliver on promises we make them.

Interviewer: *OK, so back to the customers. We look at the numbers and find that subscription is waning. How would you examine why this is happening?*

Interviewee: I'd try to look at syndicated reports first to get a sense of where people were turning for business news, but I think I'd want to go out and speak to the target customers, both subscribers and non-subscribers.



Interviewer: *So you are able to conduct a market research study and you find that those who never subscribed have very low brand awareness, and those who have lapsed and even some who still subscribe feel that the magazine is not as insightful as it once was. You also find that the average reader is now 10 years older than they were 5 years ago. What are your thoughts?*

Interviewee: Touching on the different things you told me, I'm less concerned about the low brand awareness because if we have a differentiated, quality product we can advertise it and position it better in the future. I'm more concerned that people feel the magazine is not as insightful. You said the product hasn't really changed, so does this have to do with competition?

Interviewer: *What do you think?*

Interviewee: I'd think so, maybe even to do with the availability of news on the internet. So maybe simply reporting the news doesn't get it done anymore. And that could also explain some of the change in reader demographics – younger people realize they can just go to the internet for basic news, and they get that instantly.

Interviewer: *So that was a key issue. What might Prime Inc. do about it?*

Interviewee: Maybe change the content a bit, do more human interest stories that skew younger. I guess it depends on what advertisers they want and how far they can go without angering their existing readers. Maybe do stories about how execs were using technology to solve problems in their organizations.

Interviewer: *So who is the demographic here?*

Interviewee: For a business newspaper, I'd say 35-65, mostly male but not entirely, and high income bracket.

Interviewer: *Right, and who would want to market to those people*

Interviewee: Business services, leisure, apparel, etc.

Interviewer: *So let me give you 2 pieces of information. Of the top 5 magazines doing business news, Prime Inc has the lowest share of that demographic and the highest price per ad per 1000 readers. What does that tell you?*

Interviewee: They're overcharging – if they're not hitting the target demographic (even if they're hitting other ones) why would an advertiser pay the most money to advertise with them?

Interviewer: *So what do you recommend?*

Interviewee: Lowering ad prices to be competitive.

Interviewer: *Do we want to lower them across the board? What should we be aware of?*



Interviewee: Well not all advertisers are the same. Some do more volume and perhaps different industries have different price elasticities. We should give our volume advertisers bigger discounts so we don't lose them unless they are not sensitive to price – we can probably look historically as well as currently to see trends.

Interviewer: *Great. And how about alternative strategies to grow the business to those who haven't heard of it?*

Interviewee: I'd think you can use the website, try to leverage other media within the organization, develop proprietary content, or launch a large ad campaign



Case 17: CPG Company

OVERVIEW

Firm: BCG

Round: 2006 FT, Final

Content: Qualitative and quantitative

CASE QUESTION

A large food and beverage consumer packaged goods company (CPG) is the client. They have a 40% market share in traditional CPG channels, like large format grocery. They have a 17% market share in food service. They hired us to help them grow that food service business. How would you structure that?

INTERVIEWER BRIEFING

Recommended Approach

This is really a 3 Cs case. If you use the 3 C's, develop questions around them that are relevant to the specific question, and probe the interviewer for new, relevant information, then you can crack it. This is not a numbers case. Some knowledge of marketing and consumer packaged goods would be helpful. Later in the case, using the 4 Ps to help think through the questions about how specifically to grow market share would be helpful in generating ideas.

Key Facts

There's a lot of additional information in this case. The interviewee should probe for this information because it's essential for solving the case.

- The client has one major competitor in beverage, but no major competitor in their foods portfolio. The beverage portfolio consists of sodas, waters, juices, teas, health drinks, and new specialty bottled drinks—no alcoholic beverages. The food portfolio consists primarily of snack foods—for throughout the day (ie, breakfast, lunch, and later). The client's main competitor is a major beverage company—but they don't have any sort of food portfolio.
- The client's share is 17% across the food service market. But within this market, the market share for the client varies in the different segments and subsegments. There are three primary segments in the food service market: 1) cafeteria/ workplace food service, 2) restaurants (which is comprised of fast food, slow fast food/ alternative fast food, and other), 3) entertainment/ theme park venues, and 4) misc. Amongst these segments, restaurants is definitely the largest—70% of the market.
- Within restaurants, there are a few subsegments: 1) fast food (Wendy's, BK, McDonald's), 2) slow fast food/ alternative fast food (Chipotle, Au Bon Pain, Cosi, California Pizza Kitchen Express, Wolfgang Puck's), and 3) Misc. (corner deli, mom and



pop, etc.). The fast food segment is dominated by our major competitor, who has a 70% market share in the segment. Slow fast food/ alternative fast food is a smaller segment, but it is growing at 30% a year, as a part of a trend of eating healthier.

EXAMPLE DIALOGUE

Interviewee: I'd like to make sure that I understand the case. So, our client is a traditional CPG, with a beverage and foods portfolio. Can you tell me a bit more about that portfolio? I'm assuming that they have sodas and juices? Do they have other types of drinks?

Interviewer: *Sure. The beverage portfolio consists of sodas, waters, juices, teas, health drinks, and new specialty bottled drinks—no alcoholic beverages.*

Interviewee: What about their food portfolio? What is that comprised of?

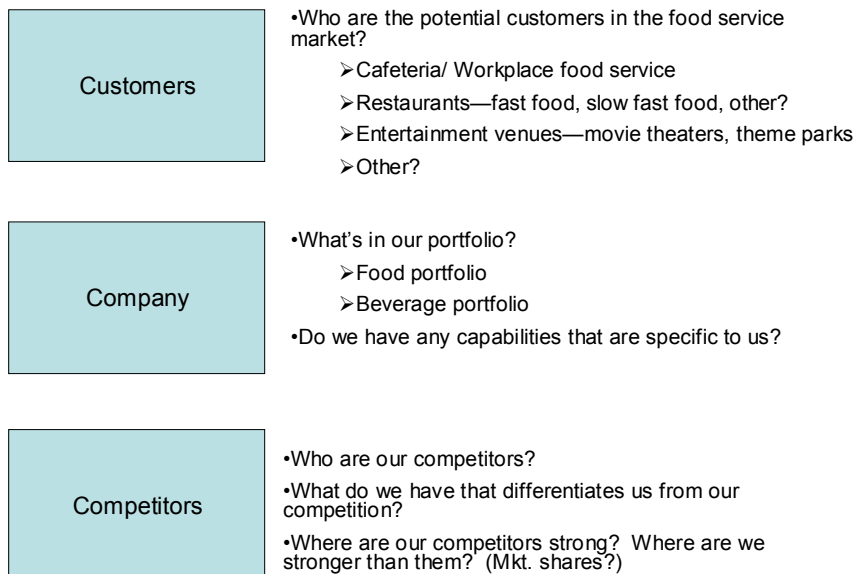
Interviewer: *The food portfolio consists primarily of snack foods—for throughout the day (ie, breakfast, lunch, and later). That's one thing that makes us distinctive from our main competitor—we have both a food and beverage portfolio. Our main competitor only has beverages.*

Interviewee: Interesting. So, they have a pretty comprehensive portfolio. Can I have a minute to structure my thoughts?

Interviewer: *Sure. Take your time.*

(Now the interviewee should take a few minutes to structure the case. This will require more than just laying out a framework but also identifying key questions that will help them drive to an answer.)

Interviewee: I would like to know about three main things: 1) customers, 2) our company and capabilities, and 3) the competition and how we can differentiate between them. In those three main categories of information, there are a few key questions that I'd like to answer. This is how I might structure it:



Interviewer: *That's good. What would you like to start with first?*

Interviewee: Well, we already talked about our company and portfolio a bit; is there anything else that I should know about our company. Like, do we have any capabilities that our competition doesn't have?

Interviewer: *Not really.*

Interviewee: Well, I'd like to know a little more about customers in the market. I don't know much about the food service market, but I imagine that there's a lot that falls into the group of customers.

Interviewer: *You're right. You've listed a few. Why don't we discuss those.*

Interviewee: Sure. I imagine that food service includes things like cafeterias, both at schools and universities. And, I know that the company that I worked at before school had a cafeteria—maybe that as well?

Interviewer: *Yep. What else do you have?*

Interviewee: Well, then, there's movie theaters and theme parks—maybe places like bowling alleys?

Interviewer: *Sure. What else?*



Interviewee: Yes, there's also restaurants. And then, there might be customers that I'm forgetting. Is there anything else?

Interviewer: *Sure, there's probably some sort of miscellaneous group—but you've covered the main ones. So, of those main three, what do you think the sizes of the segments are?*

(The Interviewer is testing the interviewee's business judgment/ common sense in asking for the interviewee's sense of the sizes of the segments.)

Interviewee: I think that miscellaneous is pretty small. And then, the entertainment one is also pretty small and stable. But the cafeteria/ workplace one, I imagine that that's a little bigger—and maybe it's growing. And then, the restaurant one, I bet that's the largest. Though, I'm not sure how fast it's growing.

Interviewer: *You're basically right. The restaurant segment is definitely the largest—it's about 70% of the food service market.*

Interviewee: If I think back to the original question of helping them grow their share in the food service segment, I think that I would focus on the restaurant segment. Can you tell me a bit more about that segment of the food service market?

Interviewer: *Well, what do you think about that segment?*

(Part of what the Interviewer is testing here is the interviewee's ability to use common sense to lead the case and see what they can deduce on their own.)

Interviewee: Well, I imagine that restaurants includes a few subsegments. For a CPG, high-end restaurants are out. But in the low-end, there's probably fast food, delis and mom and pop shops, and miscellaneous.

Interviewer: *That's right. There's one big, new one that you're missing. It's referred to as "slow fast food." It's new chains with more made-to-order, healthier food. It's restaurants like Chipotle, Boston Market, Cosi, etc. It's part of people's need to eat healthier.*

Interviewee: Interesting. I imagine that this new segment is growing a lot faster than the other segments. But it's small. If I think about the original question of the case, if I was going to try to grow market share, I might try to grow it throughout the restaurant segments. However, before I make a recommendation, I'd like to know a bit more about the competition—we haven't talked much about that.

Interviewer: *You're right. Well, the competition is pretty fierce in the fast food subsegment. E have one main competitor. They only make beverages, and they have a 70% market share in fast food.*



Interviewee: Wow. Well, given that, I would focus on the other two segments: this new slow, fast food segment and the mom and pop segment.

Interviewer: Great. What would you do to try to focus on them and grow your market share with them?

(The interviewee might want to take a few seconds to think about the answer to this question. It's kind of the crux of the answer to the case. Since it's a marketing question—how to grow market share—the 4 Ps come in as a handy framework to use to think through the case.)

Placement	•Are there customers in the restaurant segment that we're not taking care of right now? New sub-segments?
Product	•Can we offer specialty products/ tailor-made products to restaurants? •Can we offer new product packaging that is specific to our restaurant customers?
Price	•Should we compete on price? Do we want to compete on price? (Price competition could grow our unit share, but hurt of \$ volume market share—and our profits overall)
Promotion	•Can we cross-promote with our restaurant customers? •Will cross-promotion help grow restaurant customers' loyalty to us?

Interviewee: Well, there are a few things that we could to grow our share with these customers. In terms of price, well, we could change our price, but I don't think that we want to compete on price. That could help us grow our unit volume share, but it might hurt our dollar volume share and definitely would hurt our profits. In terms of placement/ distribution, is there a way that we can grow our share by distributing to new customers/ locations?

Interviewer: Maybe. But we already distribute to a majority of the market. What else could you do to grow share.

Interviewee: In terms of promotion, maybe they can create partnerships with certain customers or customer chains and run cross promotion campaigns—like advertising campaigns, or coupons. And in terms of product, maybe they can create customized products for their restaurant customers? Like new packaging? Specialty sizes, etc?



Interviewer: *Yep. That's great. That's a good way to grow share. But can you just quickly give me some of the pros and cons of specialty or custom products?*

(Here, the Interviewer is testing the interviewee's ability to identify some business risks with new ideas—testing business judgment and risk identification.)

Interviewee: Well, on the pros side, it would build a special relationship with the customer; it might differentiate us from our competition. On the cons side, our profit margins would definitely decrease on custom products. So, we might grow our share, but not our overall bottom line.

Interviewer: *That's right. So, is there anything else—other than custom products or any of the ideas that you mentioned—that can help us grow our share in the restaurant segment?*

(Here, the Interviewer is looking for a specific answer. If the interviewee doesn't get it, don't worry about it. You can still get the job without getting the answer to this one.)

Interviewee: Well, when I think about the information that I got early in the case, the only thing that I have that really differentiates us from our competition is the fact that we have a food and beverage portfolio. Maybe we can leverage the fact that we can distribute food products and beverage products to restaurant customers to grow our market share?

Interviewer: *Yes, that's right. So, how would you summarize your recommendation?*

Interviewee: Sure. The client wants to grow their market share in the food service segment. The best way to do that would be to focus on the restaurant segment of the food service market, because that's the largest part of the market—nearly 70%. Within that, the client should focus on the non-fast food segments, as those segments aren't dominated by our major competitor and are still up for grabs. And they're growing faster than the traditional fast food segment. The best ways to grow market share within those sub-segments—without hurting the bottom line—would be to leverage the breadth of our portfolio, having both food and beverage products to offer which our competition does not, and perhaps by making custom products, where it doesn't hurt our profit margins.



Case 18: Institutional Asset Manager Fees

OVERVIEW

Firm: BCG

Round: 2006 FT, Final

Content: Qualitative and quantitative

CASE QUESTION

BCG has been hired by an institutional asset manager client. The client is very profitable and doing well. However, they know that they are leaving money on the table: they know that they have a lower average percent fee of assets under management than their competition—lower than the industry average. They’ve hired BCG to tell them why that is, but in the end we told them more than that.

INTERVIEWER BRIEFING

Recommended Approach

This case dives right into data and question and answer. There’s a question that the client asked, but the interviewer hinted up front in the case question that he/she wants the interviewee to do more than just answer the client’s question. This is a good case for practicing looking at data and taking the case interview past answering the question and more towards “fixing” the business problem.

The interviewer should expect the interviewee to drive the case, but should provide the data (exhibits 1 & 2) up-front pretty quickly and make sure that the interviewee understands the industry and how the fee structure works very early on.

Key Facts

- Institutional asset managers’ revenues are based on fees from clients. Clients are charged a fee, which is structured by some percentage of assets under management. You can assume 5% if it makes the discussion easier—though, the percent fee never comes up.
- Institutional asset management companies are companies like Fidelity. They can serve any sort of large, investing clients—pension funds, CalPers, TIA-CREF, hospitals, foundations, universities, etc. They do have retail division that serve individuals with their 401Ks and mutual funds, but this case focuses on the institutional investor side of the business.

EXAMPLE DIALOGUE

Interviewer: *So, first do you have any questions?*

Interviewee: Sure, I want to make sure that I understand how the client’s business works. I don’t know too much about institutional investors or asset managers; so, I’d like to just make sure that I understand the business model before we dive into the problem.



Interviewer: *Sure, that makes sense. What can I tell you?*

Interviewee: I imagine that an institutional asset manager is a company like a Fidelity, or Capital Group Companies.

Interviewer: *That's right.*

Interviewee: And I bet that their clients can range from anyone—you and me investing our savings in a mutual fund to large investors like pension funds.

Interviewer: *Yep, that's right. Pension funds like TIA-CREF and CalPers, or foundations and universities are some of the bigger clients. We're going to focus on that part of the business—the institutional investor part of the business, not on you and me and our individual 401-Ks.*

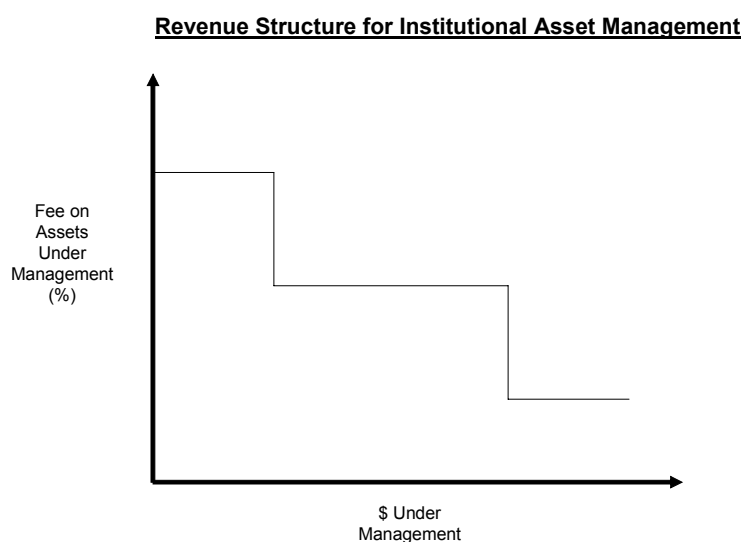
Interviewee: Great. And you said that they generate revenues from fees which are based on a percent of assets under management?

Interviewer: *Right. What's your sense of how that percent might change?*

Interviewee: My guess is that larger clients—clients with larger amounts of assets under management—might have a lower percent fee charged to them, because they still generate more revenues with a smaller percent fee.

Interviewer: *That's right. In fact, I have an exhibit that shows you what that looks like.*

Exhibit 1



It's a stepped function. What would you image the cost structure of different size clients is like?



Interviewee: Huh. Well, I don't know the business too well, but I imagine that it costs about the same to service a large client as it does a small one.

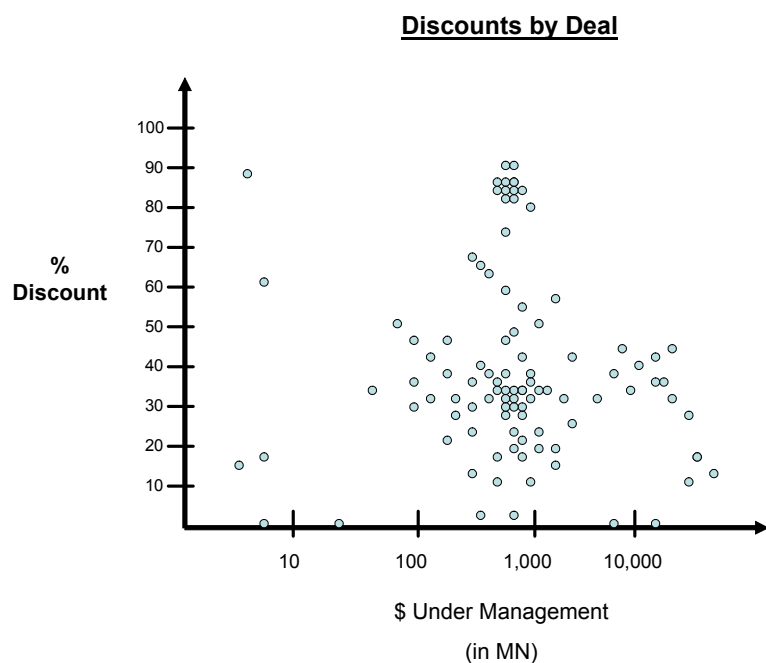
Interviewer: Yes, that's correct. So, what does that tell you?

Interviewee: That tells me that large clients—in terms of assets under management—are highly profitable because they generate more in revenues—even with smaller percentage fees—at basically the same cost as a small client. Thinking back to the original question of the case: why does the client have a lower average fee than the rest of the industry, I now want to know more about the fees that they're charging different customers.

(If the interviewee did not get these points, it's not a big deal. But it is important to show them how the fees are structured and explain that larger clients generate more revenues and cost the same as any other client—therefore, they're highly profitable clients.)

Interviewer: That's right on track. Let me show you some data that we collected on the percent fees—and discounts—that they're giving to their customers.

Exhibit 2



What does this data tell you?

Interviewee: Well, it's not what I would expect. It looks like they're giving some of the highest discounts on their fees to their mid-sized clients, not to their largest ones. This may be keeping the profitability high on their large clients, but it's probably destroying margins on their mid-



sized clients, and bringing down their overall fee average. I think this shows the answer to the client's question.

Interviewer: *Sure. Now here's another question: if you were the client, what would you want to see here?*

(Here, the interviewee should generate a few ideas. A bunch are listed—if they get one or two, that's great, and you can move on. Also, it might help the interviewee to give them the exhibit so that they can draw on it—if that makes them more comfortable or helps them at all.)

Interviewee: Well, I would want to see a few things. First, I would want to be giving the biggest discounts, or the most discounts, to my larger clients. Here, it looks like they're giving the biggest discounts to the mid-sized clients. Second, there doesn't seem to be any sort of cap on the amount of discount that can be given. They're discounting up to 90% in some cases—I would think that that would destroy all profit margins. I would expect for there to be a cap around 30% or something. Last, it all seems pretty random where the discounts are given. I wonder whether the clusters of discounts represent a certain salesman/ account executive.

Interviewer: *That's right. So, what are the key steps that you need to take to fix the problem?*

Interviewee: Well, first, you have to diagnose why the discounts are the way they are right now. So, look-up who discounted what—see whether it's salesperson-driven, or if it's driven by the type of client. Then, you would need to monitor the discounting and keep track of it. Maybe you can incent the salespeople not to give discounts?

Interviewer: *Yep. That's right: monitoring and tracking, and then put in whatever systems are necessary to get the company's average fee up to at least the industry average. Great. That's it.*

Case 19: Gumby's Pizza

OVERVIEW

Firm: Bain

Round: 2006 FT, First

Content: Qualitative and quantitative

CASE QUESTION

This is a PE/ due diligence case. Bain has been hired by a private equity firm to evaluate whether or not they should buy a mom and pop pizza chain that runs pizza restaurants primarily in college towns throughout the south. The mom and pop owners are retiring and have put the business up for sale. The chain has about 30 stores and annual revenues of about \$125 million.

In order for the private equity firm to want to buy this chain, they need to be able to double gross margin in 3-5 years. They're looking for 2-3 feasible ideas of how to do that, as they would like to sell the chain to another family owner in 3-5 years.

INTERVIEWER BRIEFING

This is a profitability case, given that the PE firm is ultimately trying to double gross margin in 3-5 years. Because of that, the framework used should be a typical profitability framework of revenues and costs. Then, the interviewer would need to help focus the interviewee on the revenues side of profitability, where the 3 C's should be used to figure out what's going on in the marketplace. Once the interviewee has suggested that they'd like to look at profitability on any specific level—store-level, town-level, customer-level, product-level, the interviewer can start to supply exhibits to drive the case from there.

EXAMPLE DIALOGUE

Interviewee: It sounds like the client wants 2-3 ideas on how to double profitability, correct.

Interviewer: Yes, that's correct.

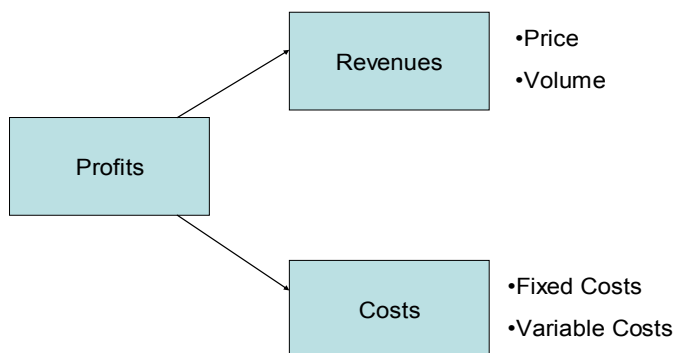
Interviewee: So, profitability is composed of revenues and costs. On the revenues side, there're two levers: price and volume. And on the costs side, there's fixed and variable costs, or depending on what kind of data we have, we could look at major line items on the balance sheet.

(Note on laying out the costs side of the case: While McKinsey recommends a costs framework of fixed vs. variable costs, Interviewer often runs down the major line items on the balance sheet: labor, materials, SG&A, PP&E. So, a respondent might want to consider using that layout to costs—it seems less “business-school formulaic” and more real world, as costs would be derived from financial statements. However, because costs are not the main thrust of this case, it's not a big point here, but a plus for a Interviewer interviewer. Also, it helps if the interviewee actually

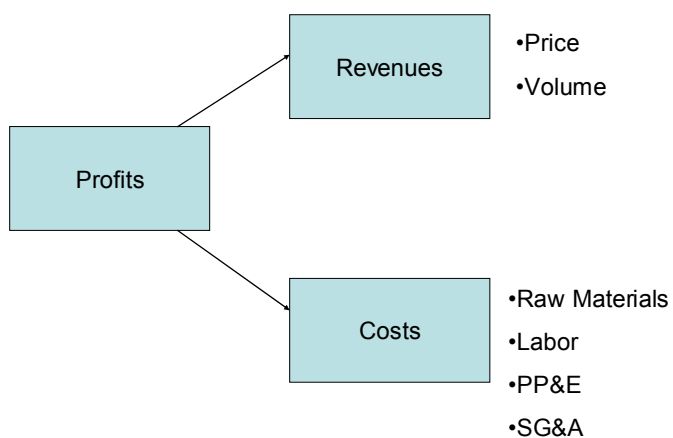


sketches out their framework on a sheet of paper in landscape format—this shows that the interviewee is thinking in “slide layout” format, which is what consulting is all about....)

One way to layout the case:



Other way:





(After laying out the profitability framework, you would need to state a hypothesis about which side—revenues or costs—you would like to focus on. It's early in the case; so, really, just make an educated guess and proceed.)

Interviewee: Because this is a private equity firm client, and they're looking to double profits in a short time period, I might focus on the costs side of the case, because it is usually easier to realize cost synergies faster than revenue synergies.

Interviewer: *Actually, while that's a good hunch about PE clients, in this particular case, there really aren't too many cost synergies to be had. It'd be better to focus on revenue gains. Given that, how might you structure your inquiry?*

(While you can use either the 3 C's or the 4P's, I think that the 3 C's work better for this case, because they provide a better overview of the marketplace and work nicely with the upcoming exhibits. But really, either would work, as they're both marketing/ revenue-growth frameworks and that's the real question being posed here.)

Interviewee: I would look at three things, and several key questions in those buckets in order to gain a sense of the market and competitive landscape and how they might be able to grow revenues. The large buckets are the following: 1) Competitive Landscape, 2) Customers and 3) Company. Here's how I might structure it, and some key questions that I'd like to ask in each bucket:

3 C's Framework with Key Questions

Competitive Landscape	Customers	Company
•National competitors? Chains? •Regional/ local competitors? •New entrants?	•What segments exist in the market? •Students? •Faculty? •Others?	•How has the company's profitability varied over time? •By store? •By region? •By product?



Interviewer: That's great. You brought up some interesting things. What kind of hypotheses do you have about the competitive landscape?

(At this point, the interviewee should use their common sense about pizza restaurants. The Interviewer interviewer is mainly trying to test their common sense and see what types of hypotheses the interviewee can generate. It's fine- probably better--for this to be more of a conversation.)

Interviewee: You mentioned that this is a mom and pop chain in southern college towns. I imagine that there are national chains that are also in those towns. I bet that those chains make the market pretty price competitive. I also wonder whether there are other local mom and pop restaurants in the towns that make competition even heavier.

Interviewer: That's great. We actually have some data on the competition.

Exhibit 1

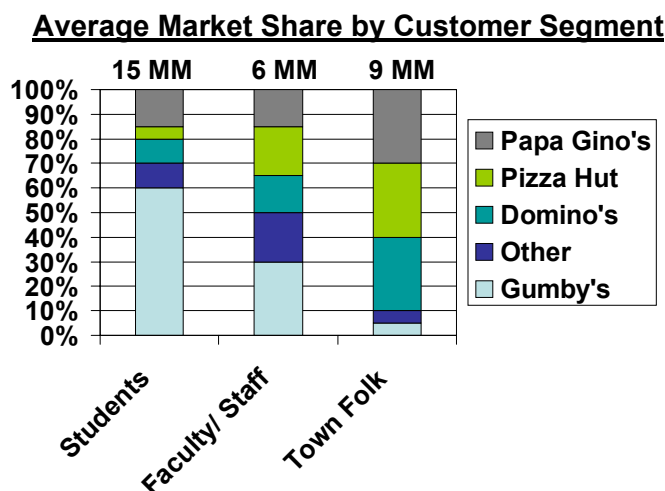
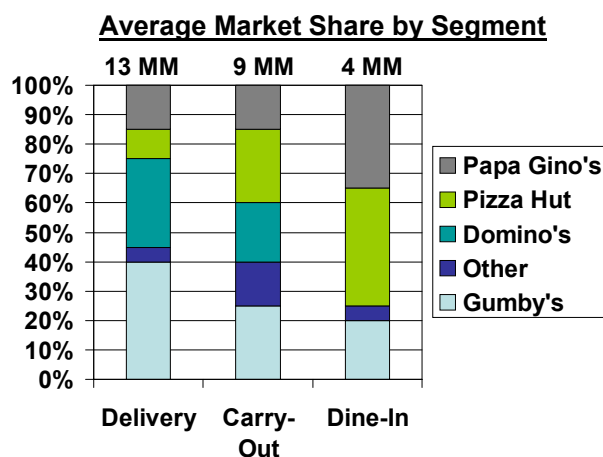


Exhibit 2





What kind of conclusions would you draw from this?

(At this point, the interviewer should give the interviewee the two exhibits. The interviewee should do a couple things. First, they should say what data is on each slide. That way, they show that they are thinking out loud, and they are sure to not miss any data on the slides. Second, they should make a summary of what the data on the slides tells them about the key question that they're trying to answer here: can the client double profits by growing the revenues side significantly in the next 3-5 years? And then, the interviewee should definitely try to synthesize at the end!)

Interviewee: Well, Exhibit 1 tells me that Gumby's has significant market share in the student customer segment—about 60%. It might be hard to grow it anymore in that segment. However, in the faculty/staff and town folk segments, there's lots of room for Gumby's to grow their share—they're down at about 30% amongst the faculty crowd and 5 % in the town folk crowd. However, the competitors in those segments look like national competitors like Dominos and Papa Gino's—I imagine that those are pretty tough competitors—particularly on price. I also would want to take into account the size of the segments in prioritizing them and trying to grow revenues in them.

Exhibit 2 shows that we're the strongest—with about a 40% market share—in the delivery segment of the market. Luckily, that's the largest segment. The dine-in segment looks like it's dominated by a couple of the national chains, and it's pretty small anyway. But the delivery and carry-out segments seem more divided, still up for grabs.

Given this new info, if I wanted to grow revenues quickly for the client—I think they have a 3-5 year time frame, I would focus on growing Gumby's share in the delivery and carry-out segments, and with the faculty/ staff and maybe the town folk.

Interviewer: *That's great. Is there any other information that you'd like to see in order to make a more refined decision on who you would target and how to grow revenues?*

(This is where the interviewer wants to see the interviewee “drilling down” on their data and hypotheses. The interviewer is also looking for the interviewee to start synthesizing and tying all the data and case together. It's always great to refer back to the original question in the case—which was profitability, not revenues.)

Interviewee: Well, the original question of the case was about profitability, not revenues. I might want to understand more about which types of segments are the most profitable before deciding where to grow my revenues. I mean, maybe the carry-out and delivery services have different profit margins—I would want to know profitability by service type, or even customer type, if that's different.

Interviewer: *Well, that's a good idea. Unfortunately, we don't have exactly the profitability data that you're looking for, but we have something close to it—exhibits 3 and 4 have some profitability data that may help you.*

(Now, the interviewer should give the interviewee the last two exhibits, and the interviewee should go through the same exercise as before, but look to synthesize and wrap up the case.)

Exhibit 3

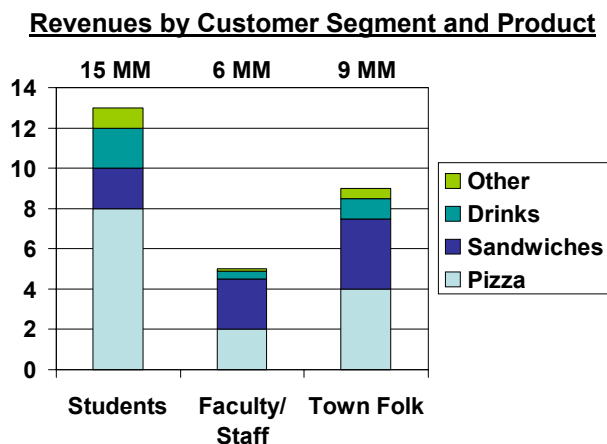
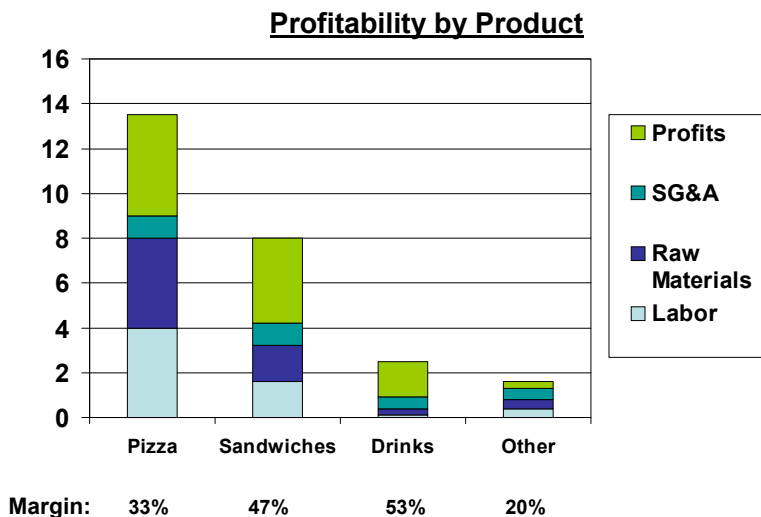


Exhibit 4



Interviewee: Well, exhibit 3 shows me the product mix by customer segment, and exhibit 4 shows the profitability by product type. Taken together, these tell me that if I do want to target the faculty/ staff and town folk to grow revenues that I really should do it by trying to get them to buy more of the more profitable products—which are sandwiches and drinks, which have profit margins of 47% and 53%.

Interviewer: Yep, that's right. Now how could you do that?



Interviewee: You could offer coupons that encourage drink-buying, or even just mention that you sell drinks when people place an order. You could offer more types of drinks—maybe people would buy more drinks if they carried a wider variety? You could advertise that you have sandwiches—revenues for sandwiches are half that of pizza, they could probably really grow that. Maybe they should change the name to “Gumby’s Pizza and Sandwiches.” Maybe they could pepper the campus with flyers about sandwiches and drinks—that would help grow the most profitable products amongst the two largest segments—students and faculty/ staff.

(The interviewee should say a lot of ideas here—doesn’t matter how great they are, but Interviewer is looking for creativity and brainstorming here.)

Interviewer: That’s enough ideas. Let’s focus on your drinks idea. Let’s say that Gumby’s currently sells each drink for \$1/ drink. Say that they want to double their gross margins through selling drinks. How many drinks would they have to sell to do that?

Interviewee: Well, at the beginning of the case, you said that their annual revenues were \$125 MN. It looks like from exhibit 4 that their gross margins right now are around 35%--is that about right?

Interviewer: Sure, 40% is about right—just to make it easier.

Interviewee: So, their current profits are 40% of \$125 MN, which is about \$50 MN. In order to double that, they would need to make \$50 MN more in profits on drinks. If the margins on drinks are 53% at \$1/ drink, then, that would mean that they would need to sell about 100 million more drinks each year. Right now, they’re selling about 3 million drinks; so, that seems really ambitious—even over a 3-5 year time horizon.

Interviewer: Right, so what does that tell you?

Interviewee: Well, the initial question in the case was: should the PE client buy Gumby’s pizza? The key question that we needed to answer was: are there 2-3 feasible ideas that could double gross margins in 3-5 years. We looked at growing profitability on the revenues side, and while it looks like there’s room for Gumby’s to grow revenues—and profits—in customer segments like faculty/ staff and town folk through highly profitable products like drinks and sandwiches, the growth that would be needed to meet the client’s time horizon and investment hurdle doesn’t seem feasible.

Interviewer: So?

Interviewee: I would recommend that they do not buy Gumby’s from the analysis that we did today.